Project Report
On

HOSTEL MANAGEMENT SYSTEM

ACKNOWLEDGEMENT

We take this occasion to thank God, almighty for blessing us with his grace and taking our Endeavour to a successful culmination. We extend our sincere and heartfelt thanks to our esteemed guide, Mrs. Preeti Saini for providing us with the right guidance and advice at the crucial junctures and for showing us the right way. We extend our sincere thanks to our respected head of the division Mrs. Preeti Saini, for allowing us to use the facilities available. We would like to thank the other faculty members also, at this occasion. Last but not the least, we would like to thank friends for the support and encouragement they have given us during the course of our work.

ABSTRACT

As the name specifies "HOSTEL MANAGEMENT SYSTEM" is software developed for managing various activities in the hostel. For the past few years the number of educational institutions is increasing rapidly. Thereby the number of hostels is also increasing for the accommodation of the students studying in this institution. And hence there is a lot of strain on the person who are running the hostel and software's are not usually used in this context. This particular project deals with the problems on managing a hostel and avoids the problems which occur when carried manually.

Identification of the drawbacks of the existing system leads to the designing of computerized system that will be compatible to the existing system with the system Which is more user friendly and more GUI oriented. We can improve the efficiency of the system, thus overcome the drawbacks of the existing system.

- Less human error
- Strength and strain of manual labor can be reduced
- High security
- Data redundancy can be avoided to some extent
- Data consistency

- Easy to handle
- Easy data updating
- Easy record keeping
- Backup data can be easily generated

INTRODUCTION

1.1 Problem definition

We have got nine hostels in our university, which consist of four boy's hostel and five girl's hostel. All these hostels at present are managed manually by the hostel office. The Registration form verification to the different data processing are done manually.

Thus there are a lot of repetitions which can be easily avoided. And hence there is a lot of strain on the person who are running the hostel and software's are not usually used in this context. This particular project deals with the problems on managing a hostel and avoids the problems which occur when carried manually

Identification of the drawbacks of the existing system leads to the designing of computerized system that will be compatible to the existing system with the system which is more user friendly and more GUI oriented. We can improve the efficiency of the system, thus overcome the drawbacks of the existing system.

1. Admin Panel

1. Admin Login

Admin can login through login form.

2. Admin Profile

Admin can manage his own profile. Admin can also change his password

3. Courses

Admin can create add course, edit courses and also delete the course

4. Rooms

Admin can create rooms and allots seater to particular rooms and assign the fees.

5. Registration
Admin can create student profile and allot the rooms.

6. Manage the Registration

Admin can manage the all the student Profile. Take a print out of all profiles and also delete the profile.

7. Complaints

Admin can manage the all the student complaints and take the appropriate action.

8. **Feedback**
Admin can view the feedback given by students for the hostel.

9. Forgot Password

Admin can also retrieve the password if admin forgot the password.

User Panel

1. **User Registration----** User can register through user registration form
2. **User Login--** User can login through login form
3. **Forgot Password—**user can retrieve password through forgot password link
4. **User Dashboard**
5. **User Profile—**User can manage own profile
6. **Book Hostel** – User can book hostel
7. **Room Details-** Booked Room Details
8. **Complaint Registration:** Student can lodge the complaint.
9. **Registered Complaint:** Student can check the registered complaint and their status.
10. **Feedback:** Student give their feedback for hostel.
11. **Change Password-** User Can change own password
12. **User access log-** User can watch last login details

SYSTEM ENVIRONMENT

2.1 Hardware Configuration

1. Pentium IV Processor

2. 512 MB RAM

3. 40GB HDD

4. 1024 * 768 Resolution Color Monitor

Note: This is not the "System Requirements".

2.2 Software Configuration

1. OS: Windows XP

2. PHP Triad (PHP5.6, MySQL, Apache, and PHPMyAdmin)

2.3 Software Features

2.3.1 PHP TRIAD

PHPTriad installs a complete working PHP/MySQL server environment on Windows platforms (9x/ NT). Installs PHP, MySQL, Apache, and PHPMyAdmin.

2.3.1.1 PHP

PHP (Hypertext Preprocessor) is a widely-used open-source server-side scripting language designed primarily for web development. It was originally created by Danish-Canadian programmer Rasmus Lerdorf in 1994 and has since been continuously developed by a large community of developers.

PHP is particularly well-suited for creating dynamic web pages and web applications. It can generate dynamic content, interact with databases, handle forms, manage sessions, create cookies, and perform many other tasks necessary for web development. PHP code is usually embedded directly into HTML documents, allowing developers to mix PHP code with HTML seamlessly.

Some key features of PHP include:

Open Source: PHP is free to use and has a large community of developers contributing to its development and support.
Cross-Platform: PHP runs on various platforms, including Windows, Linux, macOS, and many others.
Server-Side Scripting: PHP scripts are executed on the server, generating HTML which is then sent to the client's browser. This allows for dynamic content generation.
Database Integration: PHP has built-in support for working with databases like MySQL, PostgreSQL, SQLite, and others, making it easy to create database-driven web applications.
Extensive Library Support: PHP has a vast ecosystem of libraries and frameworks that provide pre-built functions and modules for common tasks, speeding up development.
Simple Syntax: PHP syntax is relatively easy to learn and understand, especially for those with a background in C-style languages like C, C++, and Java.

Overall, PHP remains one of the most popular choices for web development due to its versatility, ease of use, and extensive community support.

2.3.1.1.1 Syntax

```
<html>
<head>
   <title>PHP  Test  </title>
</head>
<body>
   <?php  echo  "<p>  Hello  World  </p>";  ?>
</body></html>
```

Note : - Code in bold letters shows the PHP code embedded within HTML

2.3.1.2 MY SQL

What is a database?

Quite simply, it's an organized collection of data. A database management system (DBMS) such as Access, FileMaker Pro, Oracle or SQL Server provides you with the software tools you need to organize that data in a flexible manner. It includes facilities to add, modify or delete data from the database, ask questions (or queries) about the data stored in the database and produce reports summarizing selected contents.

MySQL is a multithreaded,multi-user SQL database management system(DBMS). The basic program runs as a server providing multi-user access to a number of databases.Originally financed in a similar fashion to the JBoss model, MySQL was owned and sponsored by a single for-profit firm, the Swedish company MySQLAB now a subsidiary of Sun Microsystem , which holds the copyright to most of the codebase. The project's source code is available under terms of the GNU General Public Licence, as well as under a variety of proprietary agreements.

MySQL is a database.The data in MySQL is stored in database objects called tables.A table is a collections of related data entries and it consists of columns and rows.Databases are useful when storing information categorically. A company may have a database with the following tables: “Employees”, “Products”, “Customers” and “Orders”.

2.3.1.2.1 Database Tables

A database most often contains one or more tables. Each table is identified by a name (e.g. “Customers” or “Orders”). Tables contain records (rows) with data.

2.3.1.2.2 Queries

A query is a question or a request. With MySQL, we can query a database for specific information and have a record set returned.

2.3.1.2.2.1 Create a connection to a database

Before you can access data in a database, you must create a connection to the database. In PHP, this is done with the mysqli_connect() function.

Syntax

```
$con=mysqli_connect("servername", "username", "password", "db name");
```

Parameter	Description
servername	Optional. Specifies the server to connect to. Default value is "localhost:3306"
username	Optional. Specifies the username to log in with. Default value is the name of the user that owns the server process
password	Optional. Specifies the password to log in with. Default is ""

Example

In the following example we store the connection in a variable ($con) for later use in the script. The “die” part will be executed if the connection fails:

```
<?php  $con=mysqli_connect("localhost", "root", "", "acrsdb");
if(mysqli_connect_errno()){
$con=mysqli_connect("servername", "username", "password", "db name");
}?>
```

2.3.1.2.2.2 Closing a Connection

The connection will be closed automatically when the script ends. To close the connection before, use the mysqli_close() function:

```
<?php  $con=mysqli_connect("localhost", "root", "", "db_name");
if(mysqli_connect_errno()){
$con=mysqli_connect("servername", "username", "password", "db name");
}
mysqli_close();?>
```

2.3.1.2.2.3 Create a Database

The CREATE DATABASE statement is used to create a database in MySQL.

Syntax

CREATE DATABASE database_name

To get PHP to execute the statement above we must use the mysql_query() function. This function is

used to send a query or command to a MySQL connection.

2.3.1.2.2.4 Create a Table

The CREATE TABLE statement is used to create a table in MySQL

Syntax

```
CREATE TABLE table_name
    (
      column_name1
      data_type,
      column_name2
      data_type,
      column_name3
      data_type,
      ....
    )
```

2.3.1.2.3 MySQL Functions

mysqli_affected_rows — Get number of affected rows in previous MySQL

operation mysqli_change_user — Change logged in user of the active

connection mysqli_client_encoding — Returns the name of the character

set

mysqli_close — Close MySQL connection

mysqli_connect — Open a connection to a MySQL

Server mysqli_create_db — Create a MySQL

database mysqli_data_seek — Move internal result

pointer mysqli_db_name — Get result data

mysqli_db_query — Send a MySQL query

mysqli_drop_db — Drop (delete)MySQLdatabase

mysqli_errno — Returns the numerical value of the error message from previous

MySQL operation mysqli_error — Returns the text of the error message from

previous MySQL operation mysqli_escape_string — Escapes a string for use in a

mysql_query

mysqli_fetch_array — Fetch a result row as an associative array, a numeric

array, or both mysqli_fetch_assoc — Fetch a result row as an associative array mysqli_fetch_field — Get column information from a result and return

as an object mysqli_fetch_lengths — Get the length of each output in a

result mysqli_fetch_object — Fetch a result row as an

objectwsmysqli_num_rows — Get number of rows in result

mysql_pconnect — Open a persistent connection to a MySQL server

mysqli_ping — Ping a server connection or reconnect if there is

no connection mysqli_query — Send a MySQL query

mysqli_result — Get result data

mysqli_select_db — Select a MySQL

database mysqli_set_charset — Sets the

client character set mysqli_stat — Get

current system status mysqli_tablename

— Get table name of field

mysqli_thread_id — Return the current

thread ID

mysqli_unbuffered_query — Send an SQL query to MySQL, without fetching

and buffering the result (*See Appendix 2 for more My_SQL Functions.*)

2.3.1.4 phpMAdmin

phpMyAdmin is an open source tool written in PHP intended to handle the administration of MySQL over the World Wide Web. phpMyAdmin supports a wide range of operations with MySQL.Currently it can create and drop databases, create/drop/alter tables, delete/edit/add fields, execute any SQL statement, manage users and permissions, and manage keys on fields. while you still have the ability to directly execute any SQL statement. phpMyAdmin can manage a whole MySQL server (needs a super-user) as well as a single database. To accomplish the latter you'll need a properly set up MySQL user who can read/write only the desired database. It's up to you to look up the appropriate part in the MySQL manual.

phpMyAdmin can:

- browse and drop databases, tables, views, fields and indexes
- create, copy, drop, rename and alter databases, tables, fields and indexes
- maintenance server, databases and tables, with proposals on server configuration
- execute, edit and bookmark any SQL-statement, even batch-queries
- load text files into tables
- create and read dumps of tables
- export data to various formats: CSV, XML, PDF, ISO/IEC 26300 - OpenDocument Text and Spreadsheet, Word, Excel and $\LaTeX$ formats
- administer multiple servers
- manage MySQL users and privileges
- check referential integrity in MyISAM tables
- using Query-by-example (QBE), create complex queries automatically connecting required tables

- create PDF graphics of your Database layout
- search globally in a database or a subset of it
- transform stored data into any format using a set of predefined functions, like displaying BLOB-data as image or download-link
- support InnoDB tables and foreign keys
- support mysqli, the improved MySQL extension .

2.3.1.5 Apache Web server

Often referred to as simply *Apache*, a public-domain open source Web server developed by a loosely-knit group of programmers. The first version of Apache, based on the NCSA httpd Web server, was developed in 1995.

Core development of the Apache Web server is performed by a group of about 20 volunteer programmers, called the *Apache Group.* However, because the source code is freely available, anyone can adapt the server for specific needs, and there is a large public library of Apache add-ons. In many respects, development of Apache is similar to development of the Linux operating system.

The original version of Apache was written for UNIX, but there are now versions that run under OS/ 2, Windows and other platforms. The name is a tribute to the Native American Apache Indian tribe, a tribe well known for its endurance and skill in warfare. A common misunderstanding is that it was called Apache because it was developed from existing NCSA code plus various patches, hence the name *a patchy server*, or Apache server.

Apache consistently rates as the world's most popular Web server according to analyst surveys. Apache has attracted so much interest because it is full-featured, reliable, and free. Originally developed for UNIX™ operating systems, Apache has been updated to run on Windows, OS/2, and other platforms. One aspect of Apache that some site administrators find confusing — especially those unfamiliar with UNIX-style software — is its configuration scheme. Instead of using a point-and-click graphic user interface (GUI) or Windows Registry keys as most other.

Configuration Files

Apache uses a system of three text files for managing its configuration data. All three of these files (almost always) appear in Apache's ./conf directory and are designed to be edited by system administrators:

1. httpd.conf for general settings
2. srm.conf for resource settings
3. access.conf for security settings

When Apache first starts, these files are processed in the order shown above. Originally, the initial installation of Apache included default entries within each of the three files. In the most recent versions of Apache, however, the default installation has changed. Now httpd.conf is treated as the "master" configuration file and it contains all of the settings. Both srm.conf and access.conf still exist in the installation, but they contain no settings and are empty except for some comments.

Inside Httpd.conf

Traditionaly httpd.conf contained general settings such as the ServerName and Port number.These entries appear as follows in the file: ServerName compnetworking.about.com Port 80 The term "httpd" stands for *HTTP Daemon*. Recall that in a UNIX environment, the term *daemon* refers to a type of process designed to launch at system boot and continue running for very long periods of time. This file contains a number of other entries (technically called directives), but for most of these,modifications are optional. Probably the most useful of these entries is ServerAdmin.

Access and Security Settings

It is recommended practice now for Apache administrators to manage their resource and security settings from httpd.conf. Administrators of older versions of Apache can simply cut their entries from srm.conf and access.conf and paste them into the master file. If an administrator wants to go one step further and delete the two empty files, they should also place the following entries in httpd.conf to prevent Apache from attempting to access them.

SYSTEM DESIGN

3.1 Input Design

The system design is divided in to two portions. The Administrator section and the User (student's) section.

3.1.1 Administrator

1. The Administrator can allot different students to the different hostels.

2. He can vacate the students for the hostels.

3. He can control the status of the fee payement.

4. He can edit the details of the students.He can change their rooms, edit and delete the student records.

A process of converting user originated inputs to a computer-based format. Input design is an important part of development process since inaccurate input data are the most common cause of errors in data processing. Erroneous entries can be controlled by input design. It consists of developing specifications and procedures for entering data into a system and must be in simple format**.** The goal of input data design is to make data entry as easy, logical and free from errors as possible. In input data design, we design the source document that capture the data and then select the media used to enter them into the computer.

There are two major approaches for entering data in to the computer. They are

- Menus.
- Dialog Boxes.

Menus

A menu is a selection list that simplifies computer data access or entry. Instead of remembering what to enter, the user chooses from a list of options. A menu limits a user choice of response but reduce the chances for error in data entry.**Dialog Box**

Dialog boxes are windows and these windows are mainly popup, which appear in response to certain conditions that occur when a program is run. It allows the display of bitmaps and pictures. It can have various controls like buttons, text boxes, list boxes and combo boxes. Using these controls we can make a ‘dialog’ with the program.

The proposed system has three major inputs. They are Machine Registration, Machine Scheduling and Request Form.

3.2 Process Design

Process design plays an important role in project development. In order to understand the working procedure, process design is necessary. Data Flow Diagram and System Flow chart are the tools used for process design.

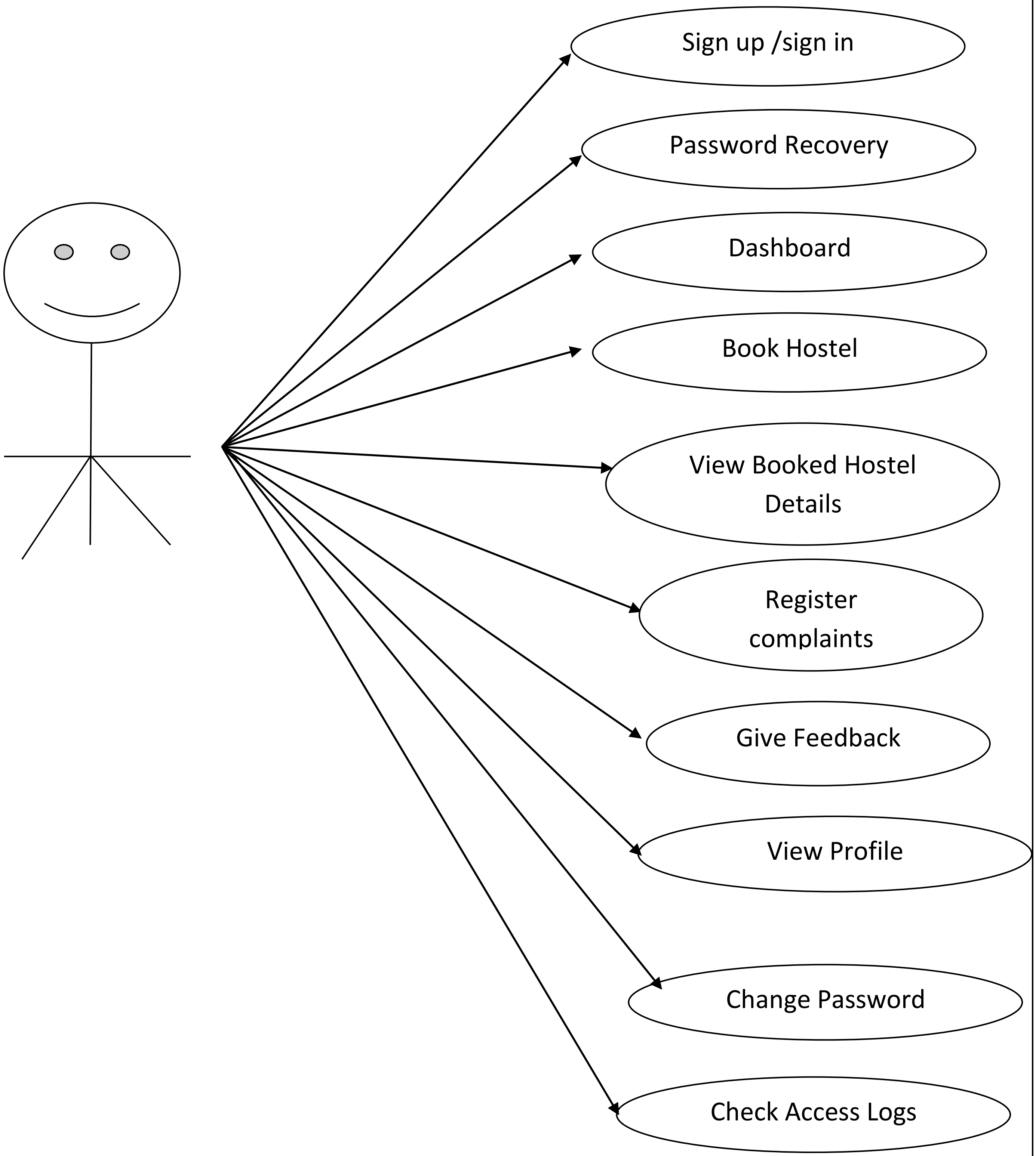
System Flow Chart is a graphical representation of the system showing the overall flow of control in processing at the job level; specifies what activities must be done to convert from a physical to logical model.

Data Flow Diagram is the logical representation of the data flow of the project. The DFD is drawn using various symbols. It has a source and a destination. The process is represented using circles and source and destination are represented using squares. The data flow is represented using arrows.

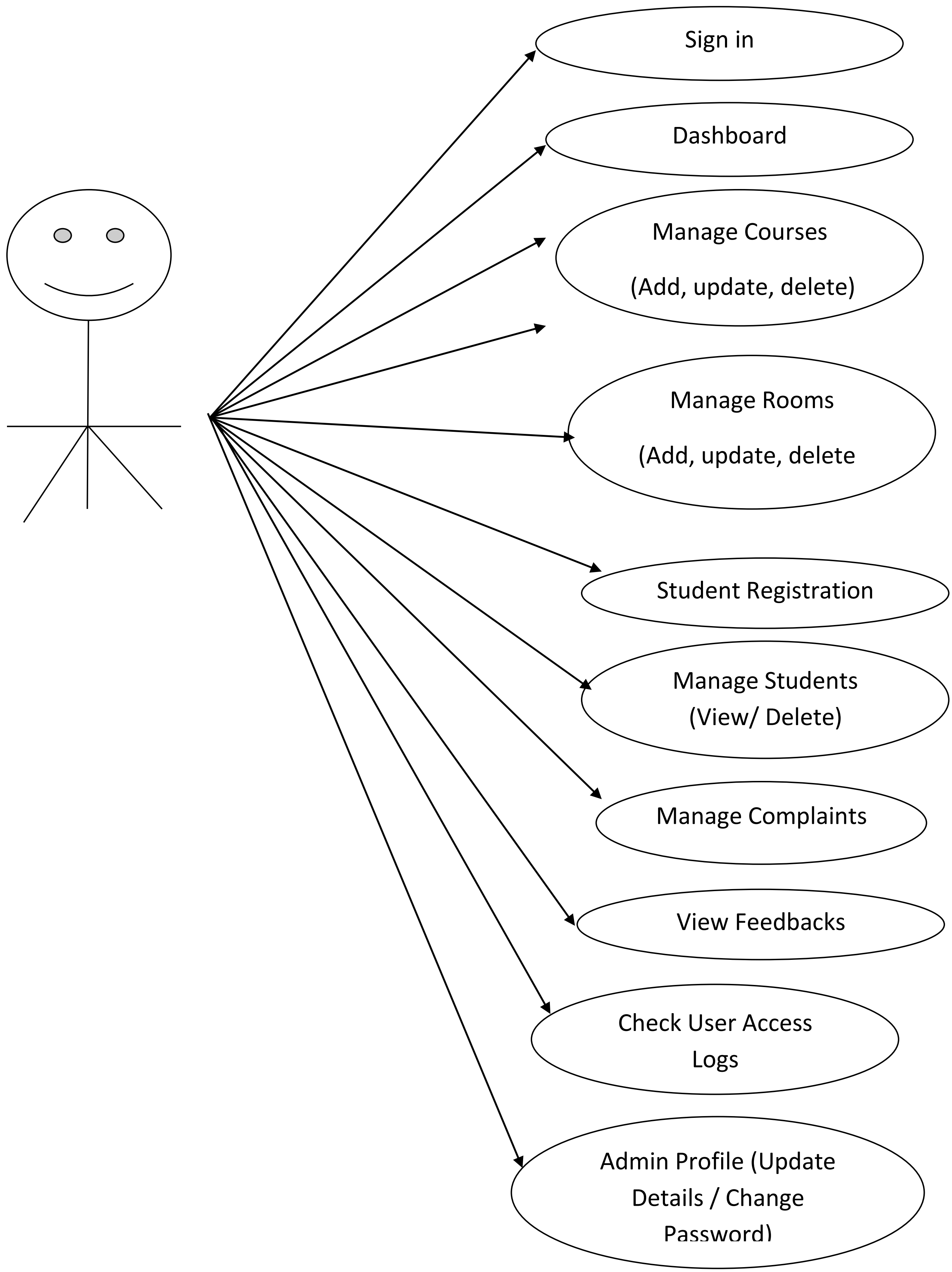
One reader can easily get the idea about the project through Data Flow Diagram.

Use case Diagram

User Use Case Diagram



Admin Use Case Diagram



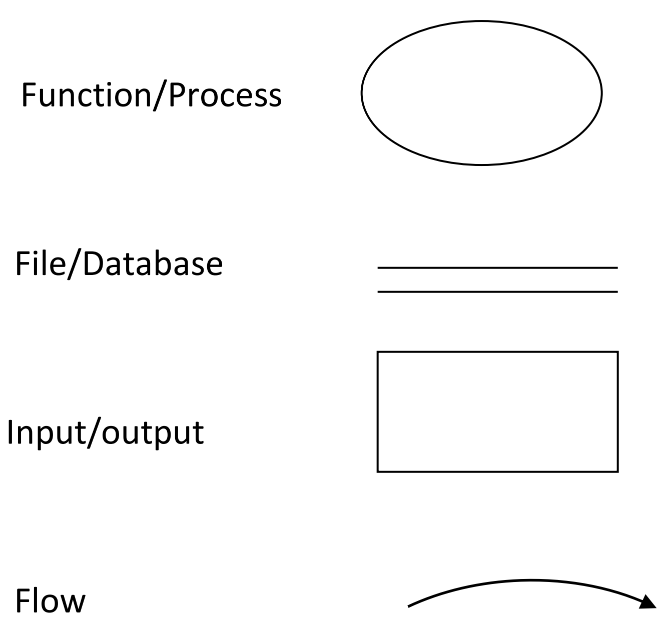
Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of the "flow" of data through an Information System. A data flow diagram can also be used for the visualization of Data Processing. It is common practice for a designer to draw a context-level DFD first which shows the interaction between the system and outside entities. This context-level DFD is then "exploded" to show more detail of the system being modeled.

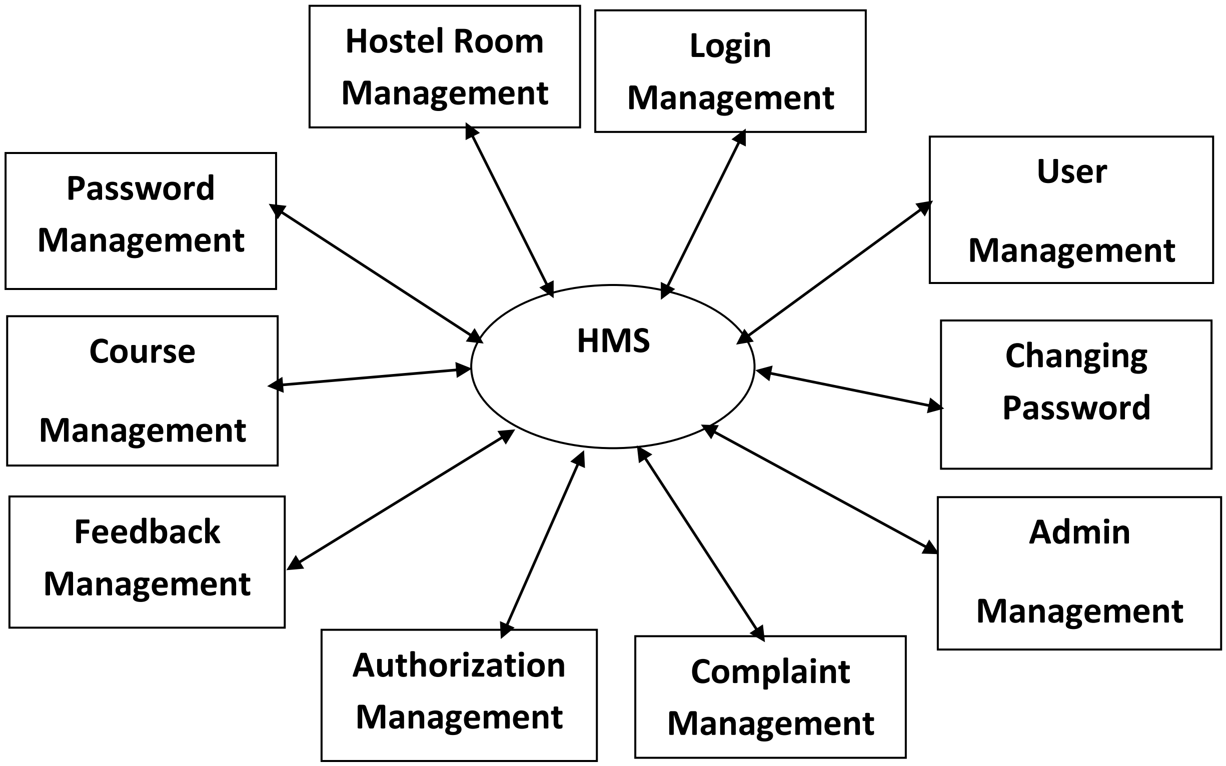
A DFD represents flow of data through a system. Data flow diagrams are commonly used during problem analysis. It views a system as a function that transforms the input into desired output. A DFD shows movement of data through the different transformations or processes in the system.

Dataflow diagrams can be used to provide the end user with a physical idea of where the data they input ultimately has an effect upon the structure of the whole system from order to dispatch to restock how any system is developed can be determined through a dataflow diagram. The appropriate register saved in database and maintained by appropriate authorities.

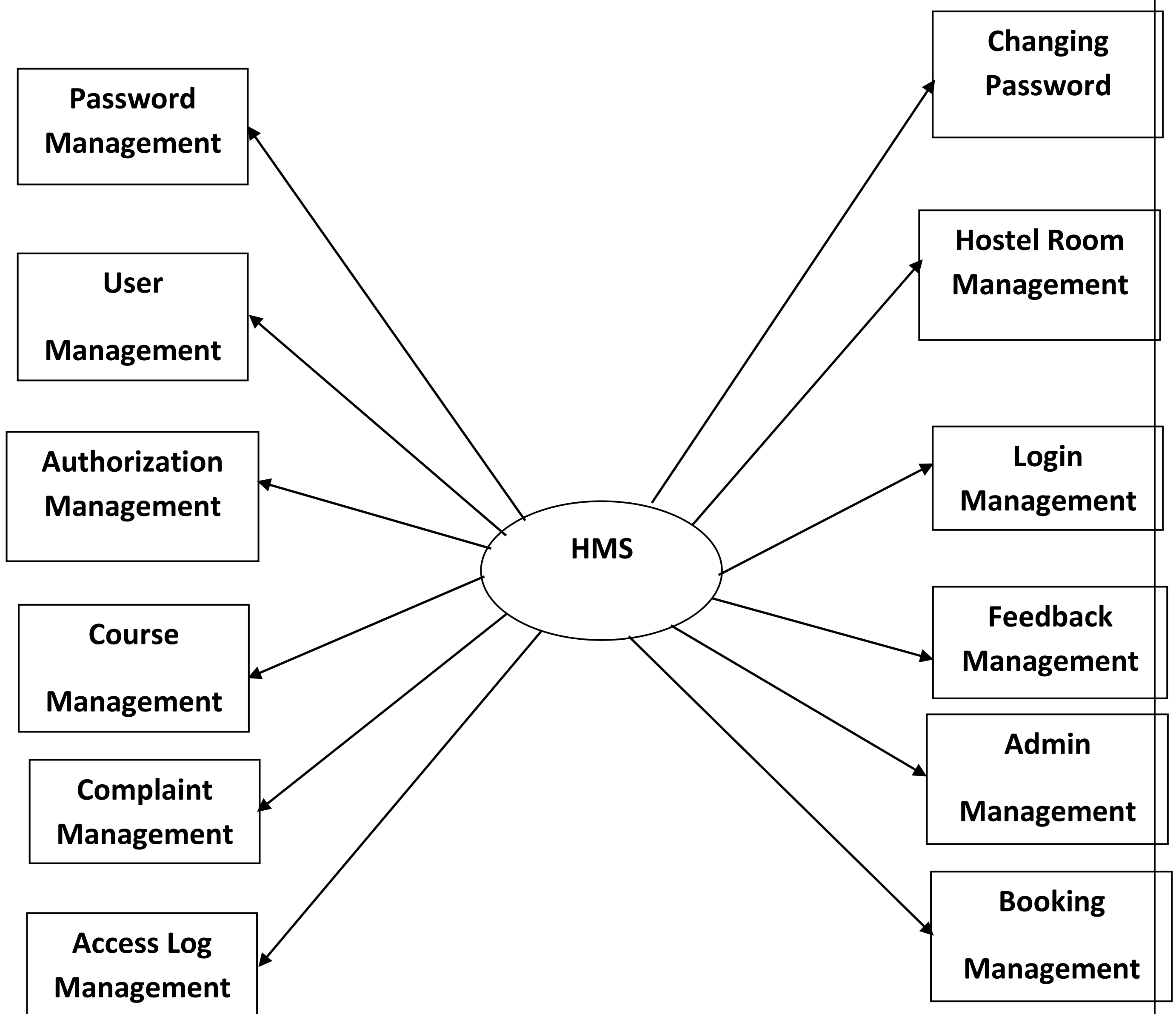
Data Flow Diagram Notation



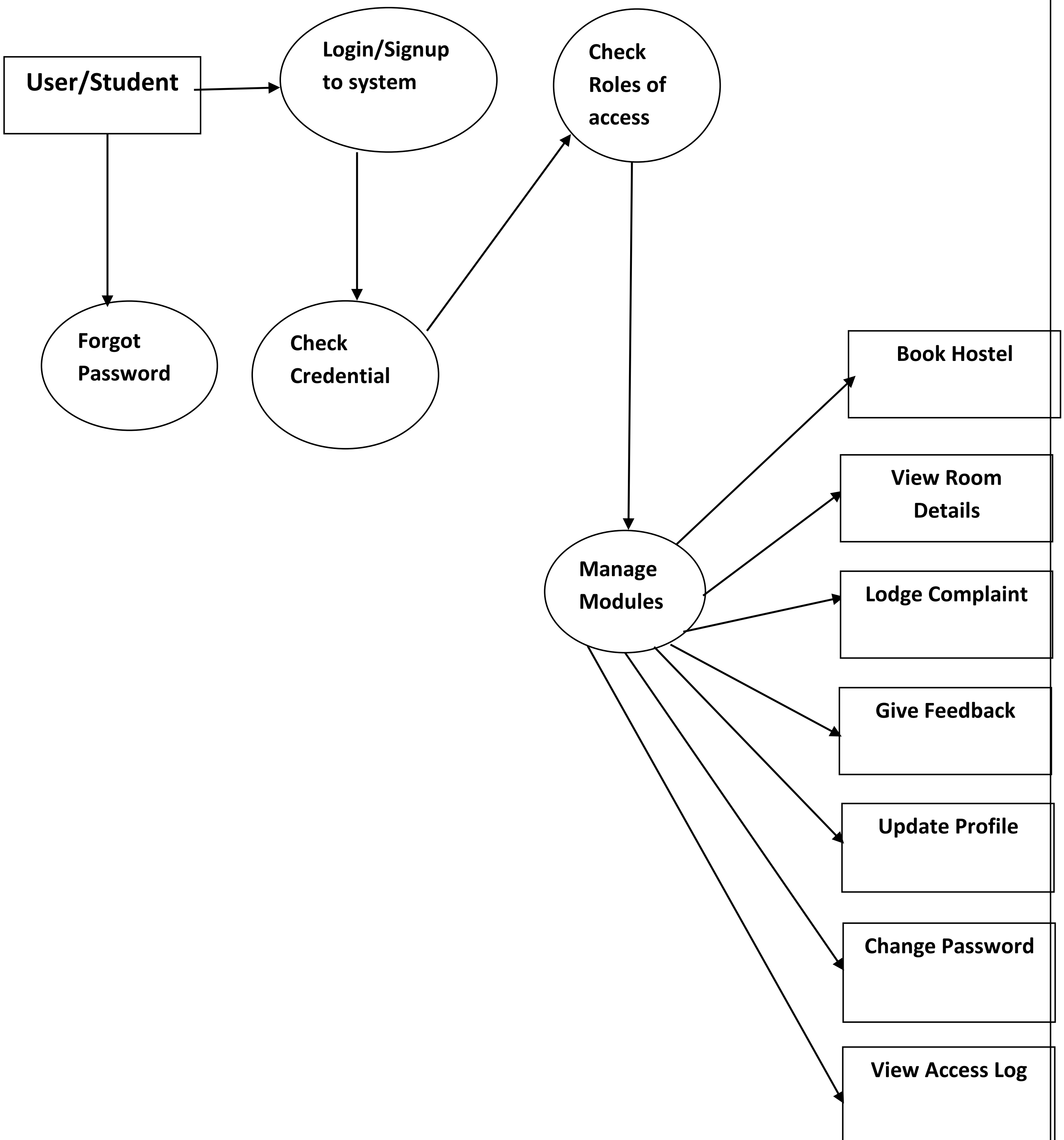
Zero Level DFD

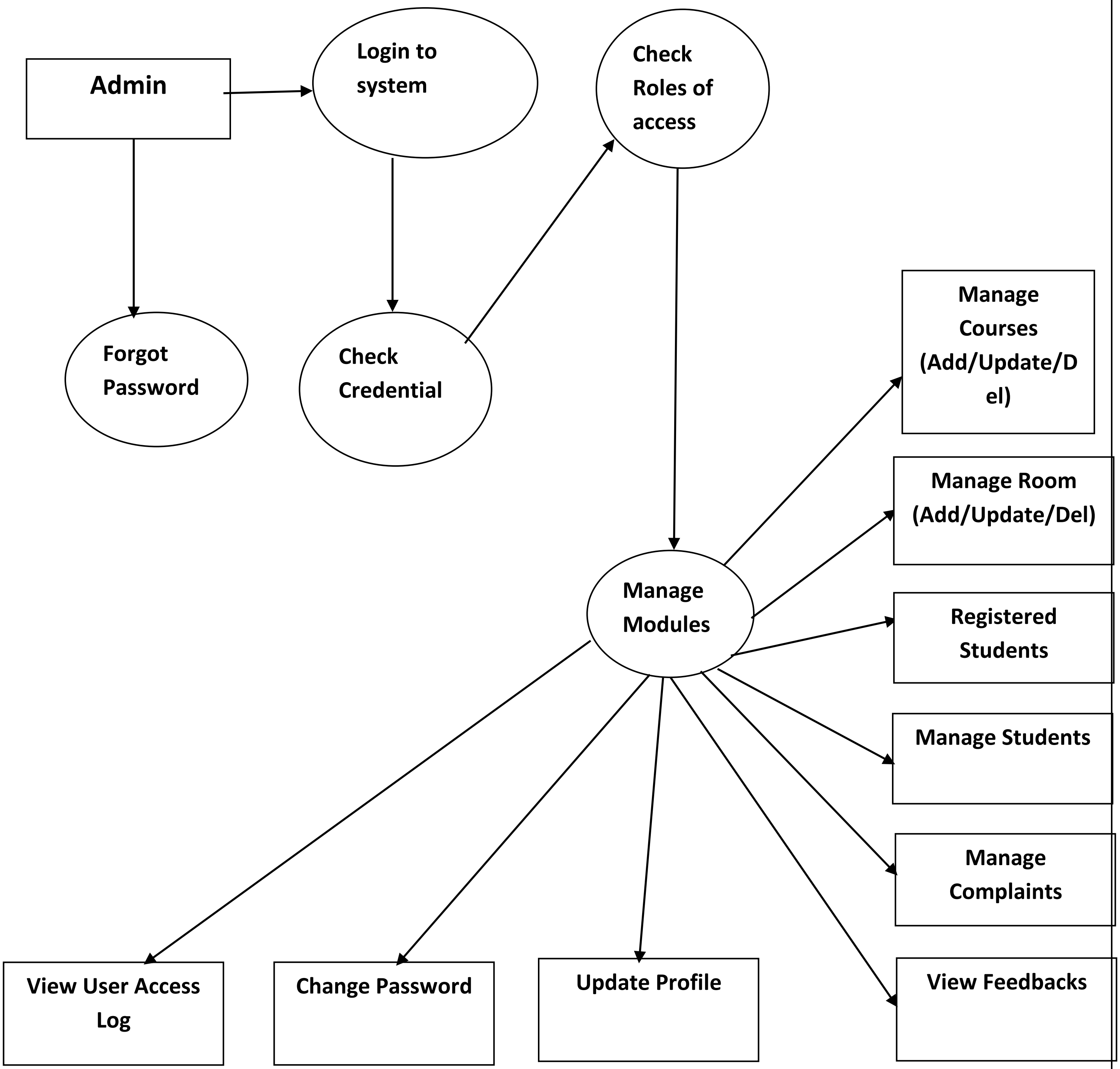


First Level DFD



Second Level DFD




Admin
Login to system
Check Roles of access
Forgot Password
Check Credential
Manage Courses (Add/Update/Del)
Manage Room (Add/Update/Del)
Manage Modules
Registered Students
Manage Students
Manage Complaints
View User Access Log
Change Password
Update Profile
View Feedbacks

ER Diagram

An Entity Relation(ER) Diagram is a specialized graphics that illustrates the interrelationship between entities in a database. ER diagrams often use symbols to represent 3 different types of information. Boxes are commonly used to represent entities. Diamonds are normally used to represent relationships and ovals are used to represent attributes.

An Entity Relationship Model (ERM), in software engineering is an abstract and conceptual representation of data. Entity Relationship modeling is a relational schema database modeling method, used to produce a type of conceptual schema or semantic data model of a system, often a relation database, and its requirements in a top-down fashion

Entity:

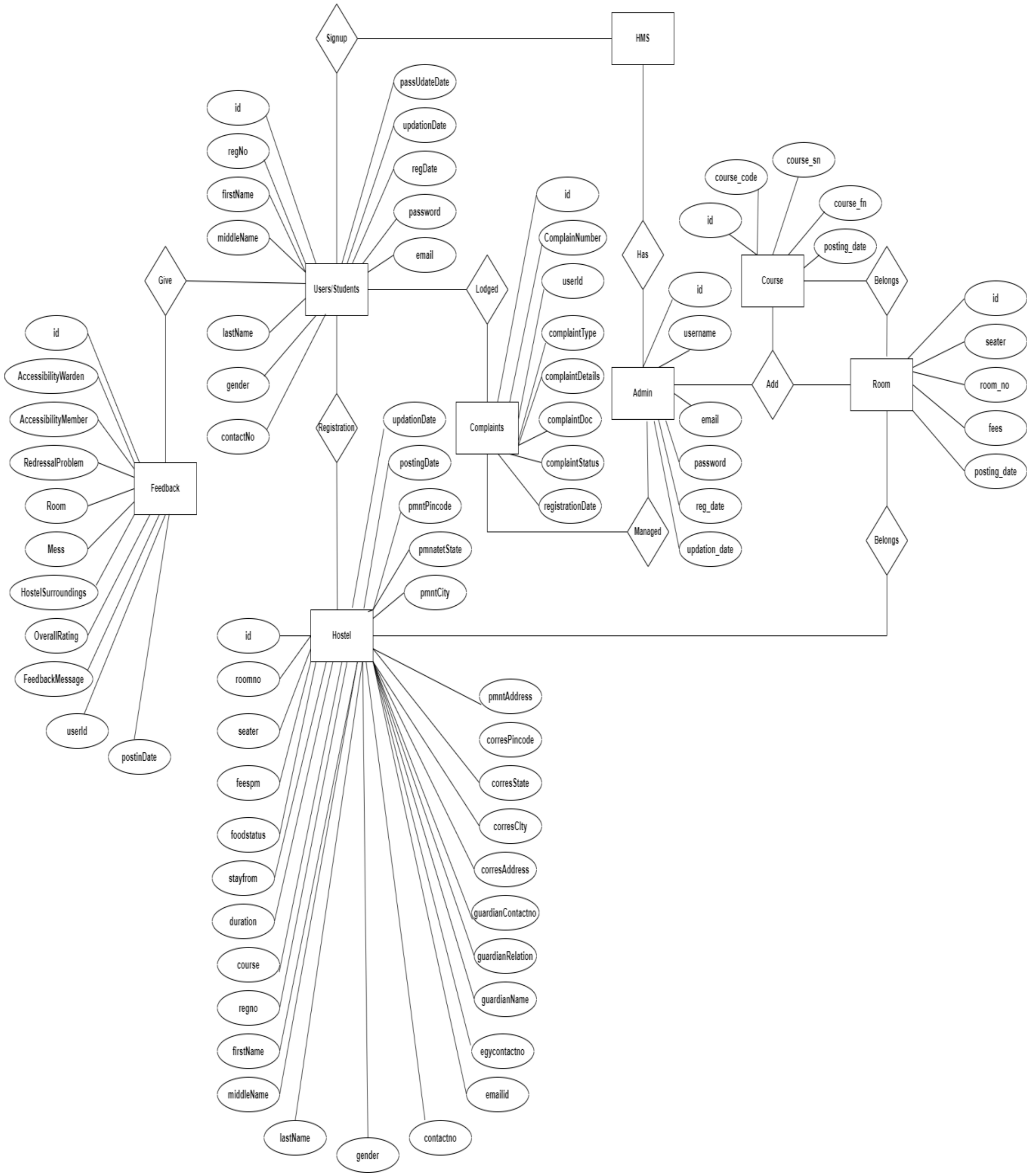
Entity is the thing which we want to store information. It is an elementary basic building block of storing information about business process. An entity represents an object defined within the information system about which you want to store information. Entities are distinct things in the enterprise.

Relationships:

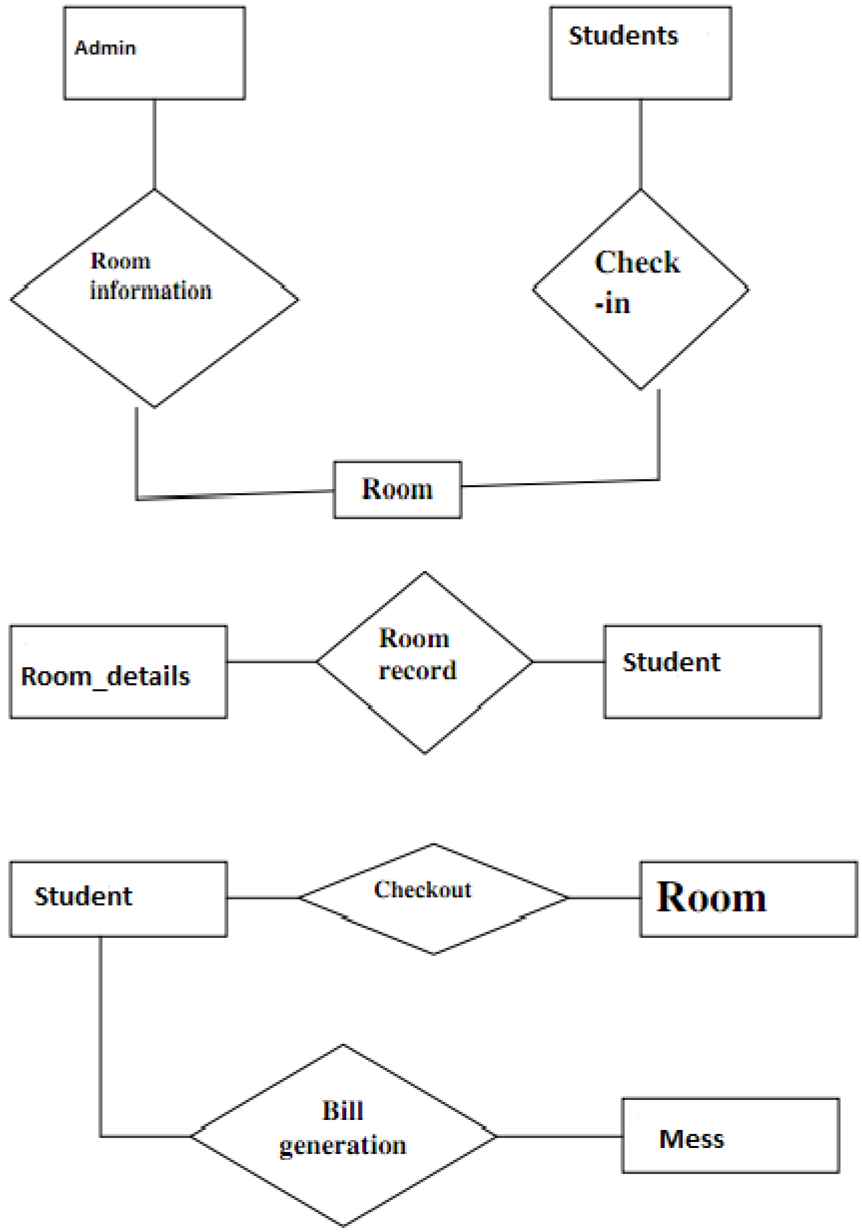
A relationship is a named collection or association between entities or used to relate two or more entities with some common attributes or meaningful interaction between the objects.

Attributes:

Attributes are the properties of the entities and relationship, Descriptor of the entity. Attributes are elementary pieces of information attached to an entity.



Signup
HMS
Users/Students
id
regNo
firstName
middleName
lastName
gender
contactNo
passUdateDate
updationDate
regDate
password
email
Give
Lodged
Has
Feedback
id
AccessibilityWarden
AccessibilityMember
RedressalProblem
Room
Mess
HostelSurroundings
OverallRating
FeedbackMessage
userId
postinDate
Registration
Complaints
id
ComplainNumber
userId
complaintType
complaintDetails
complaintDoc
complaintStatus
registrationDate
Managed
Admin
id
username
email
password
reg_date
updation_date
Course
id
course_code
course_sn
course_fn
posting_date
Belongs
Add
Room
id
seater
room_no
fees
posting_date
Belongs
Hostel
updationDate
postingDate
pmntPincode
pmnatetState
pmntCity
id
roomno
seater
feespm
foodstatus
stayfrom
duration
course
regno
firstName
middleName
lastName
gender
contactno
pmntAddress
corresPincode
corresState
corresCIty
corresAddress
guardianContactno
guardianRelation
guardianName
egycontactno
emailid



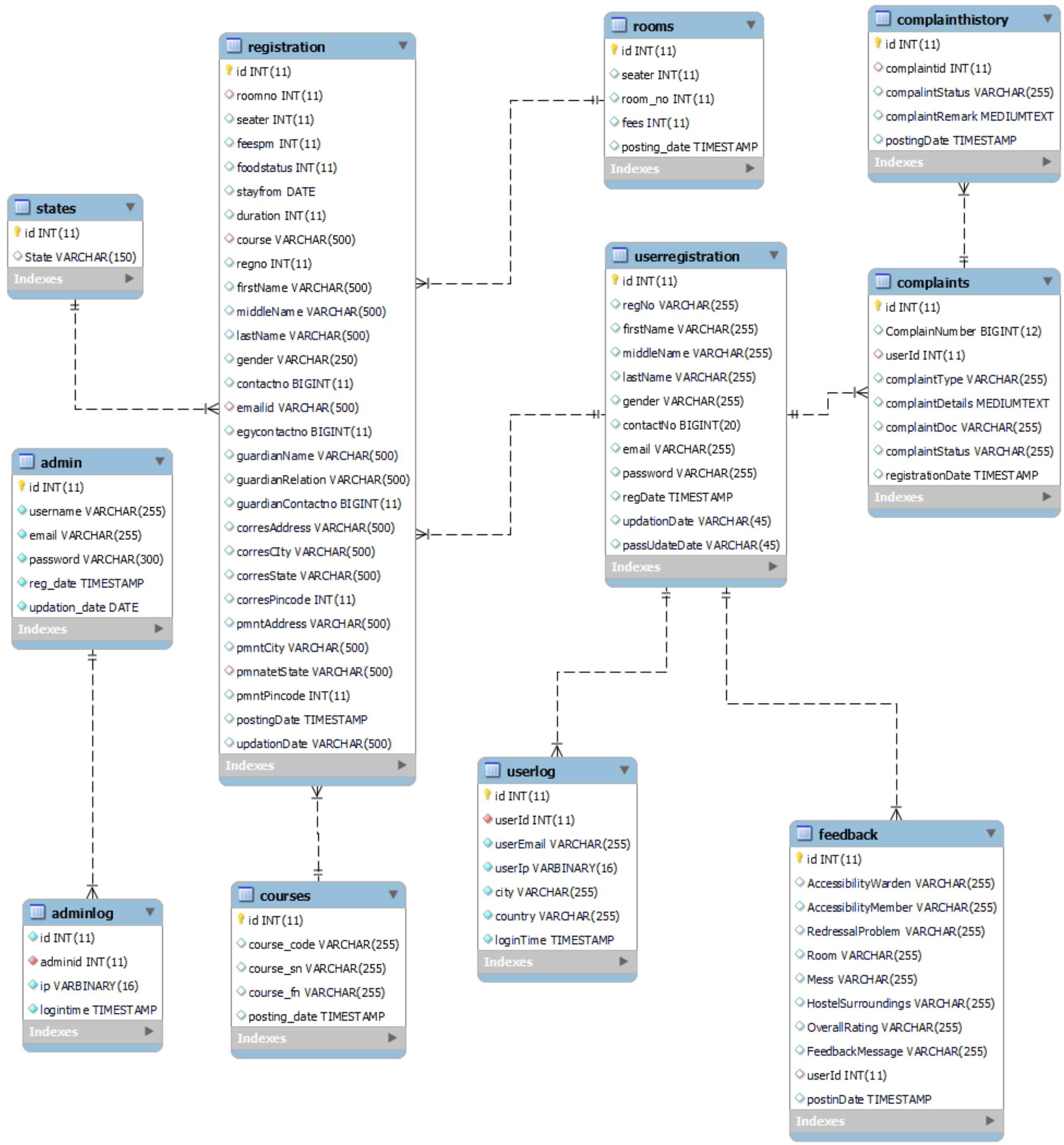
Admin
Students
Room information
Check -in
Room
Room_details
Room record
Student
Student
Checkout
Room
Bill generation
Mess

3.3 Database Design

The data in the system has to be stored and retrieved from database. Designing the database is part of system design. Data elements and data structures to be stored have been identified at analysis stage. They are structured and put together to design the data storage and retrieval system.

A database is a collection of interrelated data stored with minimum redundancy to serve many users quickly and efficiently. The general objective is to make database access easy, quick, inexpensive and flexible for the user. Relationships are established between the data items and unnecessary data items are removed. Normalization is done to get an internal consistency of data and to have minimum redundancy and maximum stability. This ensures minimizing data storage required, minimizing chances of data inconsistencies and optimizing for updates. The MySQL database has been chosen for developing the relevant databases.

Database Table Relationship



registration
id INT(11)
roomno INT(11)
seater INT(11)
feespm INT(11)
foodstatus INT(11)
stayfrom DATE
duration INT(11)
course VARCHAR(500)
regno INT(11)
firstName VARCHAR(500)
middleName VARCHAR(500)
lastName VARCHAR(500)
gender VARCHAR(250)
contactno BIGINT(11)
emailid VARCHAR(500)
egycontactno BIGINT(11)
guardianName VARCHAR(500)
guardianRelation VARCHAR(500)
guardianContactno BIGINT(11)
corresAddress VARCHAR(500)
corresCIty VARCHAR(500)
corresState VARCHAR(500)
corresPincode INT(11)
pmntAddress VARCHAR(500)
pmntCity VARCHAR(500)
pmnatetState VARCHAR(500)
pmntPincode INT(11)
postingDate TIMESTAMP
updationDate VARCHAR(500)
Indexes
rooms
id INT(11)
seater INT(11)
room_no INT(11)
fees INT(11)
posting_date TIMESTAMP
Indexes
complainthistory
id INT(11)
complaintid INT(11)
compalintStatus VARCHAR(255)
complaintRemark MEDIUMTEXT
postingDate TIMESTAMP
Indexes
states
id INT(11)
State VARCHAR(150)
Indexes
userregistration
id INT(11)
regNo VARCHAR(255)
firstName VARCHAR(255)
middleName VARCHAR(255)
lastName VARCHAR(255)
gender VARCHAR(255)
contactNo BIGINT(20)
email VARCHAR(255)
password VARCHAR(255)
regDate TIMESTAMP
updationDate VARCHAR(45)
passUdateDate VARCHAR(45)
Indexes
complaints
id INT(11)
ComplainNumber BIGINT(12)
userId INT(11)
complaintType VARCHAR(255)
complaintDetails MEDIUMTEXT
complaintDoc VARCHAR(255)
complaintStatus VARCHAR(255)
registrationDate TIMESTAMP
Indexes
admin
id INT(11)
username VARCHAR(255)
email VARCHAR(255)
password VARCHAR(300)
reg_date TIMESTAMP
updation_date DATE
Indexes
userlog
id INT(11)
userId INT(11)
userEmail VARCHAR(255)
userIp VARBINARY(16)
city VARCHAR(255)
country VARCHAR(255)
loginTime TIMESTAMP
Indexes
feedback
id INT(11)
AccessibilityWarden VARCHAR(255)
AccessibilityMember VARCHAR(255)
RedressalProblem VARCHAR(255)
Room VARCHAR(255)
Mess VARCHAR(255)
HostelSurroundings VARCHAR(255)
OverallRating VARCHAR(255)
FeedbackMessage VARCHAR(255)
userId INT(11)
postinDate TIMESTAMP
Indexes
adminlog
id INT(11)
adminid INT(11)
ip VARBINARY(16)
logintime TIMESTAMP
Indexes
courses
id INT(11)
course_code VARCHAR(255)
course_sn VARCHAR(255)
course_fn VARCHAR(255)
posting_date TIMESTAMP
Indexes

Admin Table Structure

admin

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INCREMENT
2	**username**	varchar(255)	latin1_swedish_ci		No	*None*		
3	**email**	varchar(255)	latin1_swedish_ci		No	*None*		
4	**password**	varchar(300)	latin1_swedish_ci		No	*None*		
5	**reg_date**	timestamp			No	current_timestamp()		
6	**updation_date**	date			No	*None*		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	1	A	No	

Admin Log Table Structure

adminlog

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		
2	**adminid**	int(11)			No	*None*		
3	**ip**	varbinary(16)			No	*None*		
4	**logintime**	timestamp			No	current_timestamp()		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
adminid	BTREE	No	No	adminid	0	A	No	

Complaints Table Structure

complaints

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INCREM
2	**ComplainNumber**	bigint(12)			Yes	*NULL*		
3	**userId**	int(11)			Yes	*NULL*		
4	**complaintType**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
5	**complaintDetails**	mediumtext	utf8mb4_general_ci		Yes	*NULL*		
6	**complaintDoc**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
7	**complaintStatus**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
8	**registrationDate**	timestamp			Yes	current_timestamp()		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	7	A	No	
useriid	BTREE	No	No	userId	7	A	Yes	

<u>**Complaint History Table Structure**</u>

complainthistory

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INCREM
2	**complaintid**	int(11)			Yes	*NULL*		
3	**compalintStatus**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
4	**complaintRemark**	mediumtext	utf8mb4_general_ci		Yes	*NULL*		
5	**postingDate**	timestamp			Yes	current_timestamp()		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	10	A	No	
cid	BTREE	No	No	complaintid	10	A	Yes	

<u>**Courses Table Structure**</u>

courses

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INCREMENT
2	**course_code**	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
3	**course_sn**	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
4	**course_fn**	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
5	**posting_date**	timestamp			Yes	current_timestamp()		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	7	A	No	
course_fn	BTREE	No	No	course_fn	7	A	Yes	

Feedback Table Structure

feedback

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INC
2	**AccessibilityWarden**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
3	**AccessibilityMember**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
4	**RedressalProblem**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
5	**Room**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
6	**Mess**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
7	**HostelSurroundings**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
8	**OverallRating**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
9	**FeedbackMessage**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*		
10	**userId**	int(11)			Yes	*NULL*		
11	**postinDate**	timestamp			Yes	current_timestamp()		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	4	A	No	
userId	BTREE	No	No	userId	4	A	Yes	

Room Registration Table Structure

registration

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(11)			No	*None*		AUTO_INCREI
2	roomno	int(11)			Yes	*NULL*		
3	seater	int(11)			Yes	*NULL*		
4	feespm	int(11)			Yes	*NULL*		
5	foodstatus	int(11)			Yes	*NULL*		
6	stayfrom	date			Yes	*NULL*		
7	duration	int(11)			Yes	*NULL*		
8	course	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
9	regno	int(11)			Yes	*NULL*		
10	firstName	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
11	middleName	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
12	lastName	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
13	gender	varchar(250)	latin1_swedish_ci		Yes	*NULL*		
14	contactno	bigint(11)			Yes	*NULL*		
15	emailid	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
16	egycontactno	bigint(11)			Yes	*NULL*		
17	guardianName	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
18	guardianRelation	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
19	guardianContactno	bigint(11)			Yes	*NULL*		
20	corresAddress	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
21	corresCIty	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
22	corresState	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
23	corresPincode	int(11)			Yes	*NULL*		
24	pmntAddress	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
25	pmntCity	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
26	pmnatetState	varchar(500)	latin1_swedish_ci		Yes	*NULL*		
27	pmntPincode	int(11)			Yes	*NULL*		
28	postingDate	timestamp			Yes	current_timestamp()		
29	updationDate	varchar(500)	latin1_swedish_ci		Yes	*NULL*		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	5	A	No	
emaild	BTREE	No	No	emailid	5	A	Yes	
rid	BTREE	No	No	roomno	5	A	Yes	
coursename	BTREE	No	No	course	5	A	Yes	
state	BTREE	No	No	pmnatetState	5	A	Yes	

Rooms Table Structure

rooms

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INCREMENT
2	**seater**	int(11)			Yes	*NULL*		
3	**room_no**	int(11)			Yes	*NULL*		
4	**fees**	int(11)			Yes	*NULL*		
5	**posting_date**	timestamp			Yes	current_timestamp()		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	6	A	No	
room_no	BTREE	No	No	room_no	6	A	Yes	

State Table Structure

states

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INCREMENT
2	**State**	varchar(150)	latin1_swedish_ci		Yes	*NULL*		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	36	A	No	
State	BTREE	No	No	State	36	A	Yes	

User Registration Table Structure

userregistration

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(11)			No	*None*		AUTO_INCREMEN
2	regNo	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
3	firstName	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
4	middleName	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
5	lastName	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
6	gender	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
7	contactNo	bigint(20)			Yes	*NULL*		
8	email	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
9	password	varchar(255)	latin1_swedish_ci		Yes	*NULL*		
10	regDate	timestamp			Yes	current_timestamp()		
11	updationDate	varchar(45)	latin1_swedish_ci		Yes	*NULL*		
12	passUdateDate	varchar(45)	latin1_swedish_ci		Yes	*NULL*		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	4	A	No	
email	BTREE	No	No	email	4	A	Yes	

User Logs Table Structure

userlog

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(11)			No	*None*		AUTO_INCREMENT
2	userId	int(11)			No	*None*		
3	userEmail	varchar(255)	latin1_swedish_ci		No	*None*		
4	userIp	varbinary(16)			No	*None*		
5	city	varchar(255)	latin1_swedish_ci		No	*None*		
6	country	varchar(255)	latin1_swedish_ci		No	*None*		
7	loginTime	timestamp			No	current_timestamp()		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	9	A	No	
uid	BTREE	No	No	userId	9	A	No	

3.4 Output Design

Designing computer output should proceed in an organized, well throughout manner; the right output element is designed so that people will find the system whether or executed. When we design an output we must identify the specific output that is needed to meet the system. The usefulness of the new system is evaluated on the basis of their output.

Once the output requirements are determined, the system designer can decide what to include in the system and how to structure it so that the require output can be produced. For the proposed software, it is necessary that the output reports be compatible in format with the existing reports. The output must be concerned to the overall performance and the system's working, as it should. It consists of developing specifications and procedures for data preparation, those steps necessary to put the inputs and the desired output, ie maximum user friendly. Proper messages and appropriate directions can control errors committed by users.

The output design is the key to the success of any system. Output is the key between the user and the sensor. The output must be concerned to the system's working, as it should.

Output design consists of displaying specifications and procedures as data presentation. User never left with the confusion as to what is happening without appropriate error and acknowledges message being received. Even an unknown person can operate the system without knowing anything about the system.

SYSTEM ANALYSIS

4.1 Existing System

For the past few years the number of educational institutions are increasing rapidly. Thereby the number of hostels are also increasing for the accommodation of the students studying in this institution. And hence there is a lot of strain on the person who are running the hostel and software's are not usually used in this context. This particular project deals with the problems on managing a hostel and avoids the problems which occur when carried manually

Identification of the drawbacks of the existing system leads to the designing of computerized system that will be compatible to the existing system with the system which is more user friendly and more GUI oriented. We can improve the efficiency of the system, thus overcome the following drawbacks of the existing system.

- more human error.
- more strength and strain of manual labour needed
- Repetition of the same procedures.
- low security
- Data redundancy
- difficult to handle
- difficult to update data
- record keeping is difficult
- Backup data can be easily generated

SYSTEM TESTING

System testing is the stage of implementation, which is aimed at ensuring that the system works accurately and efficiently before live operation commences. Testing is the process of executing the program with the intent of finding errors and missing operations and also a complete verification to determine whether the objectives are met and the user requirements are satisfied. The ultimate aim is quality assurance.

Tests are carried out and the results are compared with the expected document. In the case of erroneous results, debugging is done. Using detailed testing strategies a test plan is carried out on each module. The various tests performed in "**Network Backup System**" are unit testing, integration testing and user acceptance testing.

5.1 Unit Testing

The software units in a system are modules and routines that are assembled and integrated to perform a specific function. Unit testing focuses first on modules, independently of one another, to locate errors. This enables, to detect errors in coding and logic that are contained within each module. This testing includes entering data and ascertaining if the value matches to the type and size supported by java. The various controls are tested to ensure that each performs its action as required.

5.2 Integration Testing

Data can be lost across any interface, one module can have an adverse effect on another, sub functions when combined, may not produce the desired major functions. Integration testing is a systematic testing to discover errors associated within the interface. The objective is to take unit tested modules and build a program structure. All the modules are combined and tested as a whole. Here the Server module and Client module options are integrated and tested. This testing provides the assurance that the application is well integrated functional unit with smooth transition of data.

5.3 User Acceptance Testing

User acceptance of a system is the key factor for the success of any system. The system under consideration is tested for user acceptance by constantly keeping in touch with the system users at time of developing and making changes whenever required.

IMPLEMENTATION

Implementation is the stage in the project where the theoretical design is turned into a working system and is giving confidence on the new system for the users that it will work efficiently and effectively. It involves careful planning, investigation of the current system and its constraints on implementation, design of methods to achieve the changeover, an evaluation of change over methods. Apart from planning major task of preparing the implementation are education and training of users. The implementation process begins with preparing a plan for the implementation of the system. According to this plan, the activities are to be carried out, discussions made regarding the equipment and resources and the additional equipment has to be acquired to implement the new system. In network backup system no additional resources are needed.

Implementation is the final and the most important phase. The most critical stage in achieving a successful new system is giving the users confidence that the new system will work and be effective. The system can be implemented only after thorough testing is done and if it is found to be working according to the specification. This method also offers the greatest security since the old system can take over if the errors are found or inability to handle certain type of transactions while using the new system.

7.1 User Training

After the system is implemented successfully, training of the user is one of the most important subtasks of the developer. For this purpose user manuals are prepared and handled over to the user to operate the developed system. Thus the users are trained to operate the developed system. Both the hardware and software securities are made to run the developed systems successfully in future. In order to put new application system into use, the following activities were taken care of:

- Preparation of user and system documentation
- Conducting user training with demo and hands on
- Test run for some period to ensure smooth switching over the system .The users are trained to use the newly developed functions. User manuals describing the procedures for using the functions listed on menu are circulated to all the users. It is confirmed that the system is implemented up to users need and expectations.

7.2 Security and Maintenance

Maintenance involves the software industry captive, typing up system resources .It means restoring something to its original condition. Maintenance follows conversion to the extent that changes are necessary to maintain satisfactory operations relative to changes in the user's environment. Maintenance often includes minor enhancements or corrections to problems that surface in the system's operation. Maintenance is also done based on fixing the problems reported, changing the interface with other software or hardware enhancing the software.

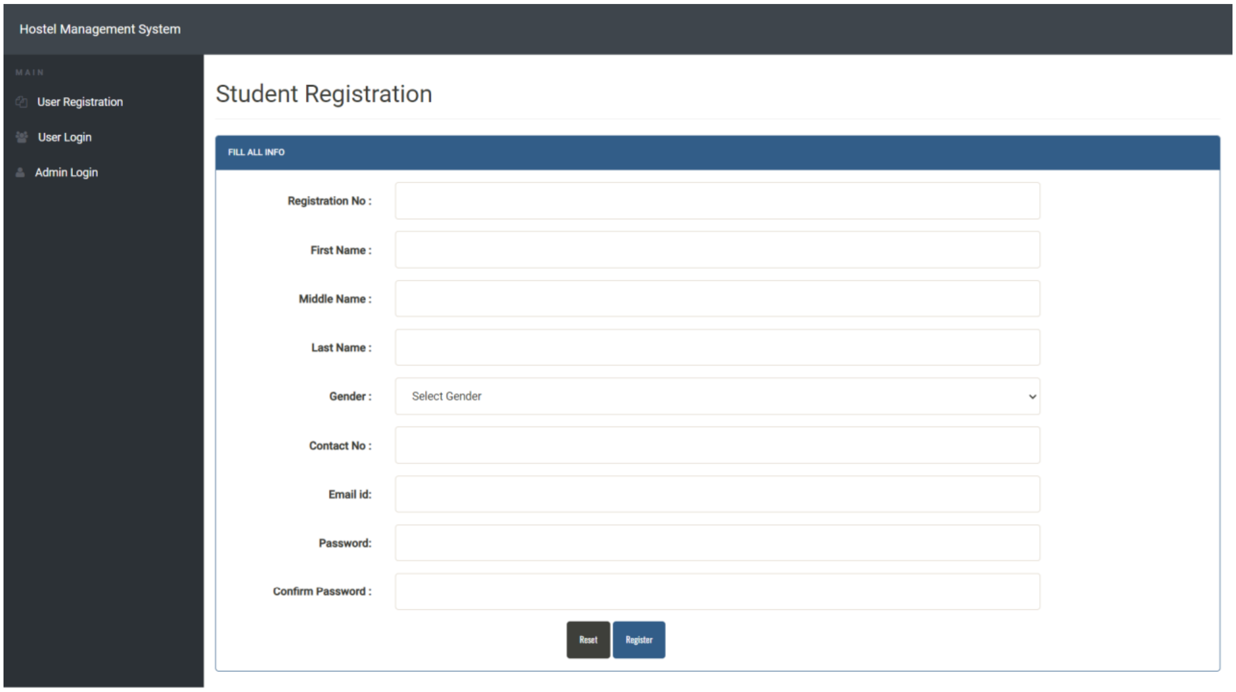
Any system developed should be secured and protected against possible hazards. Security measures are provided to prevent unauthorized access of the database at various levels. An uninterrupted power supply should be so that the power failure or voltage fluctuations will not erase the data in the files.

Password protection and simple procedures to prevent the unauthorized access are provided to the users .The system allows the user to enter the system only through proper user name and password.

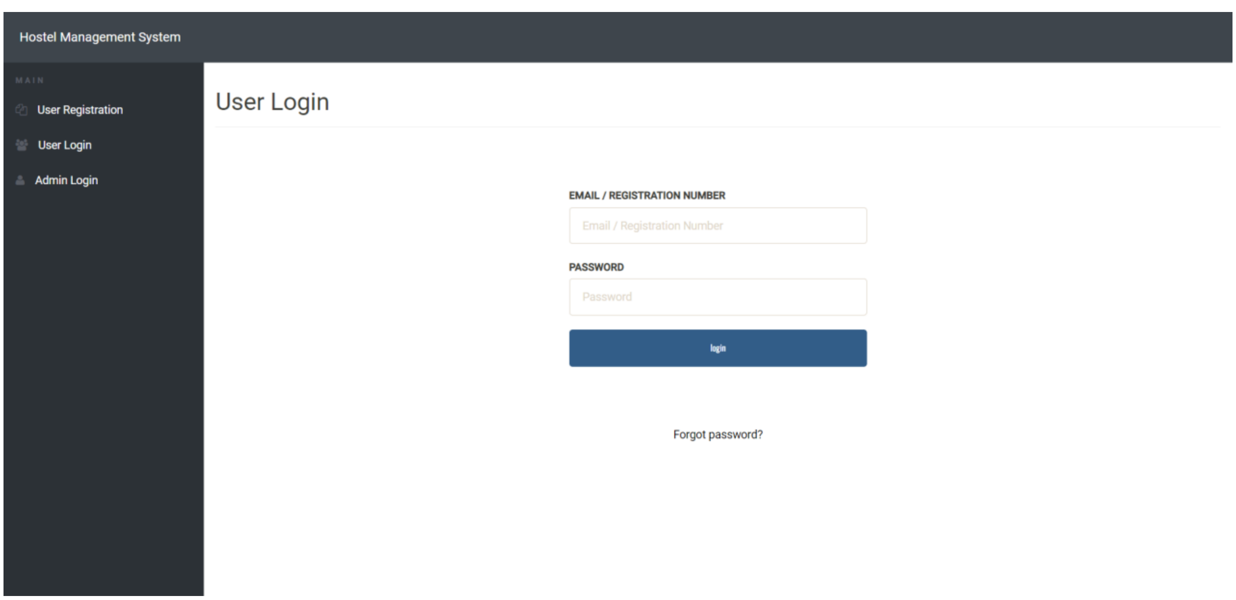
OUTPUT SCREENS

Student Panel

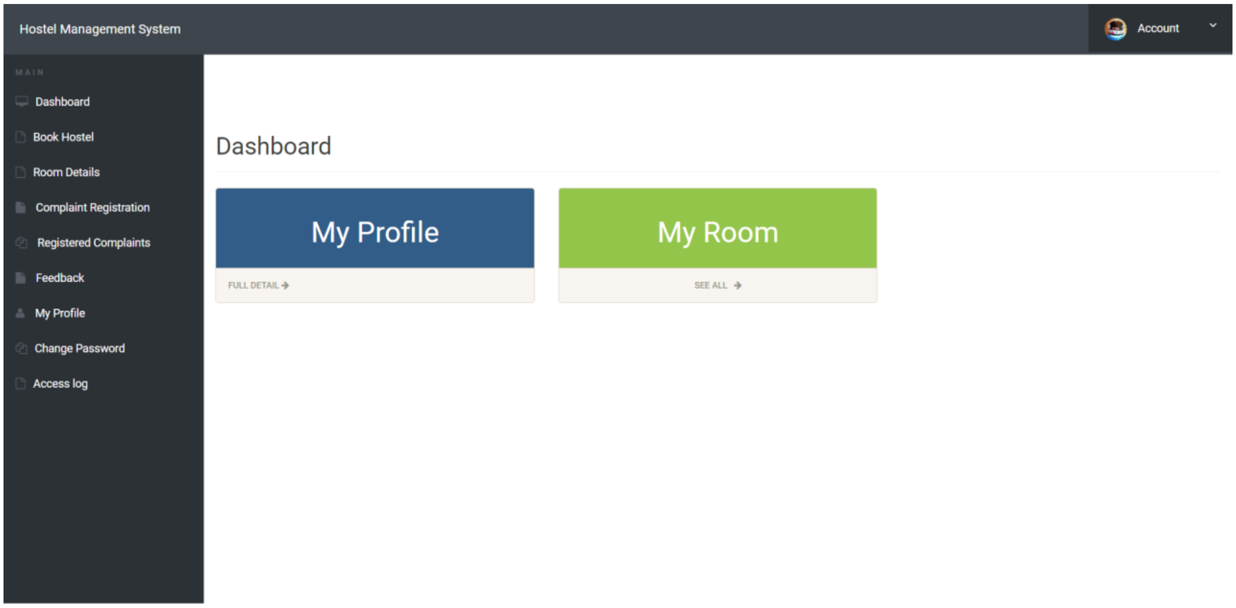
Student Registrations



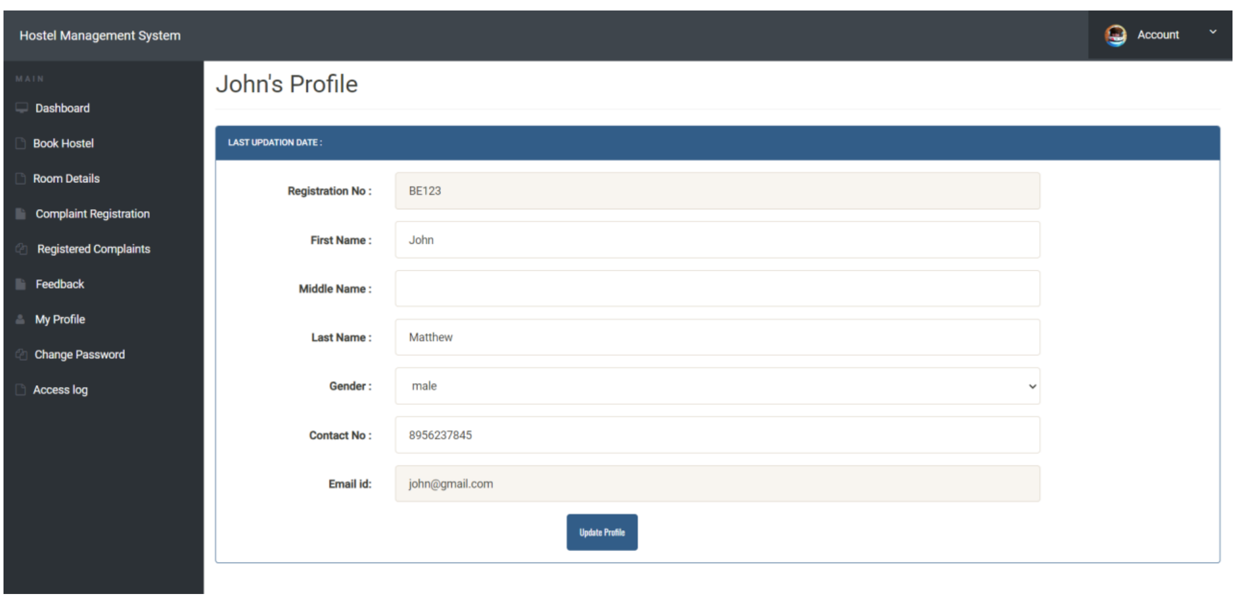
User Login



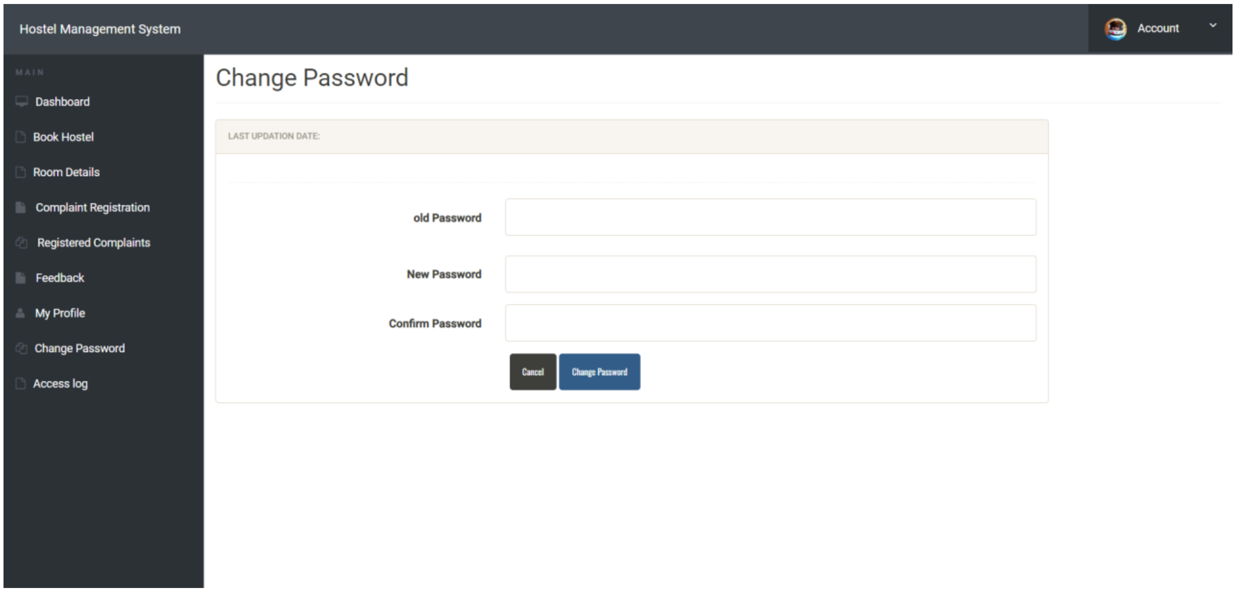
Dashboard



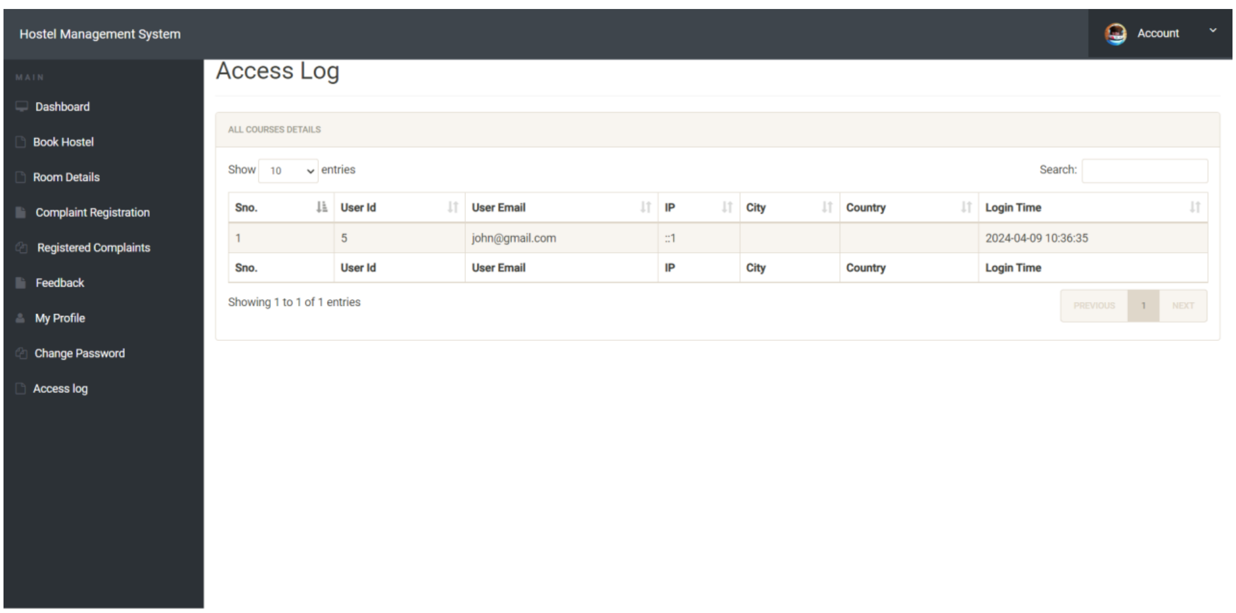
Profile



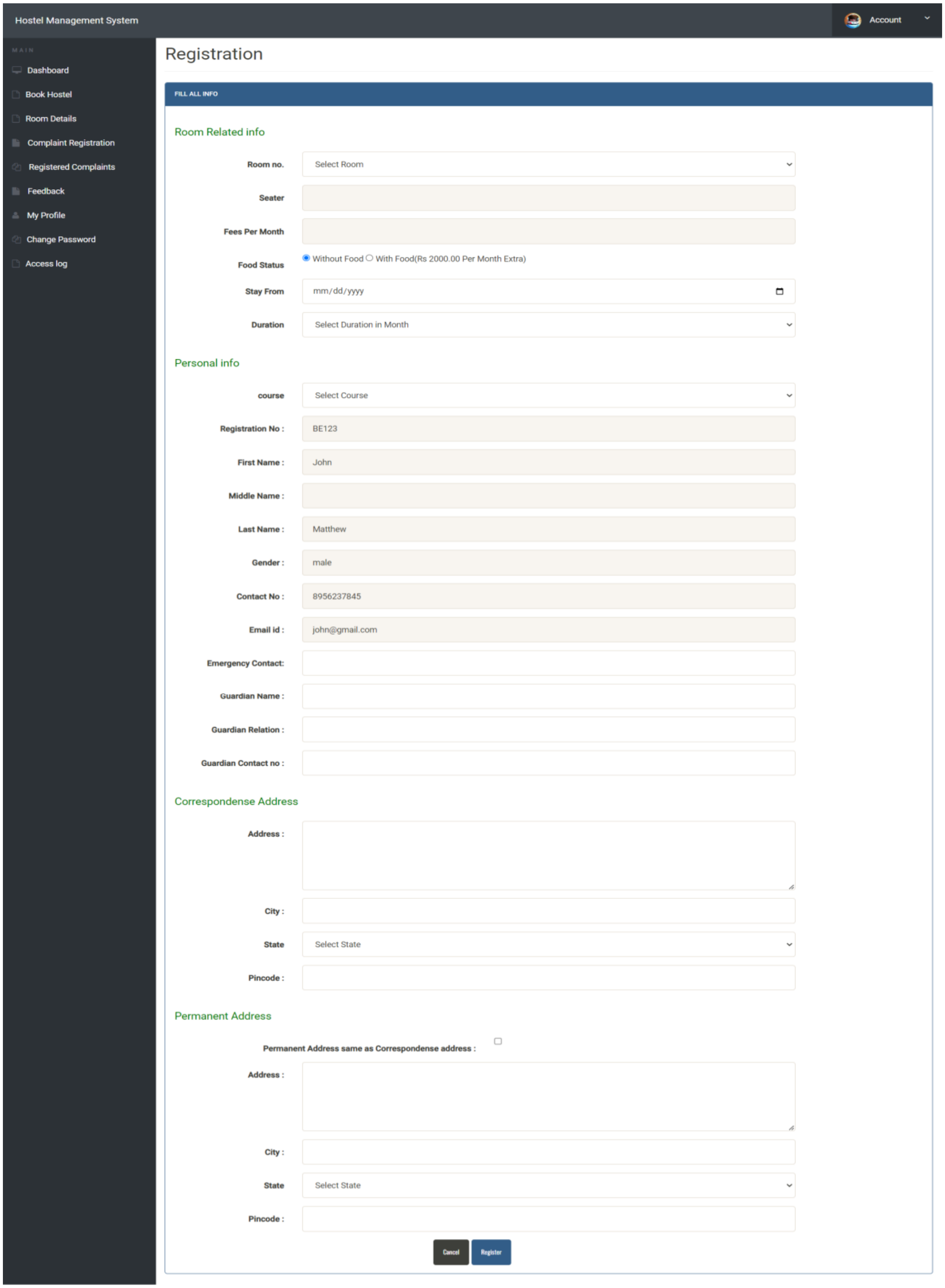
Change Password



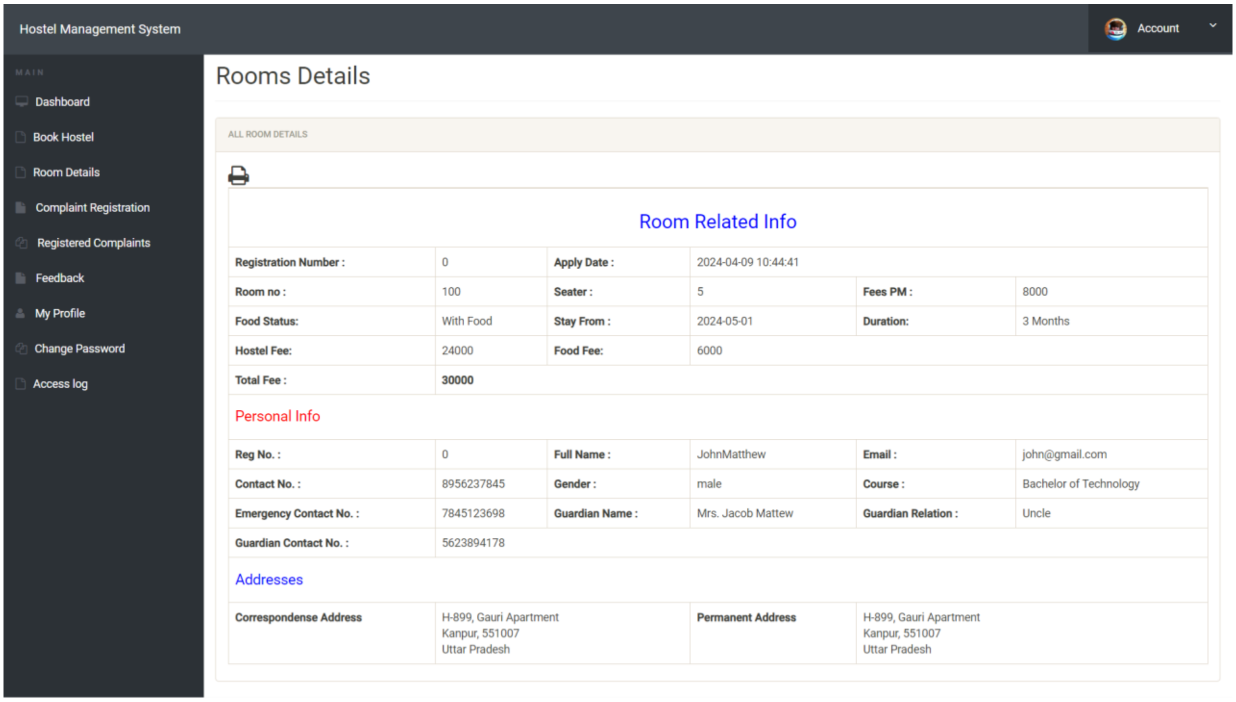
Access Log



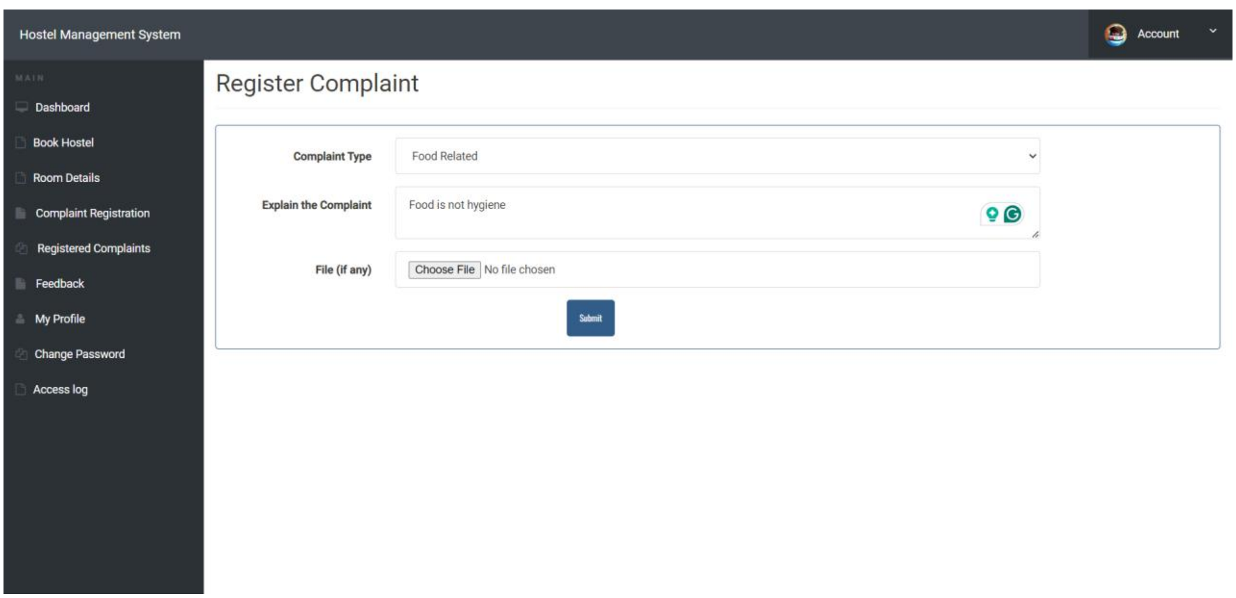
Hostel Booking Form



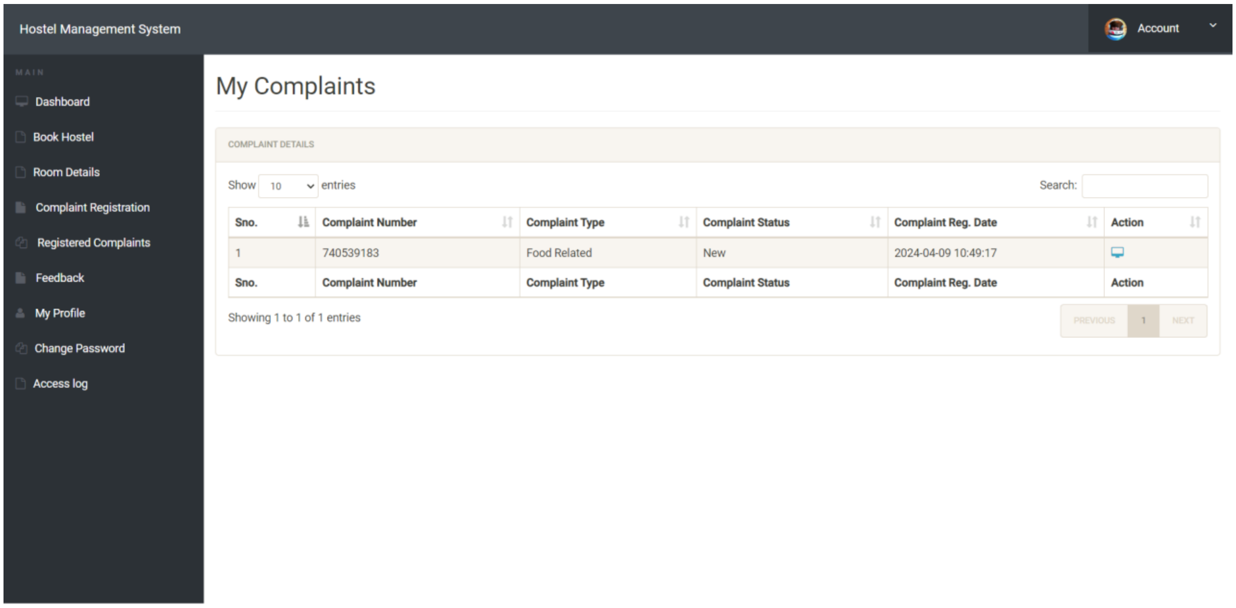
Room Details



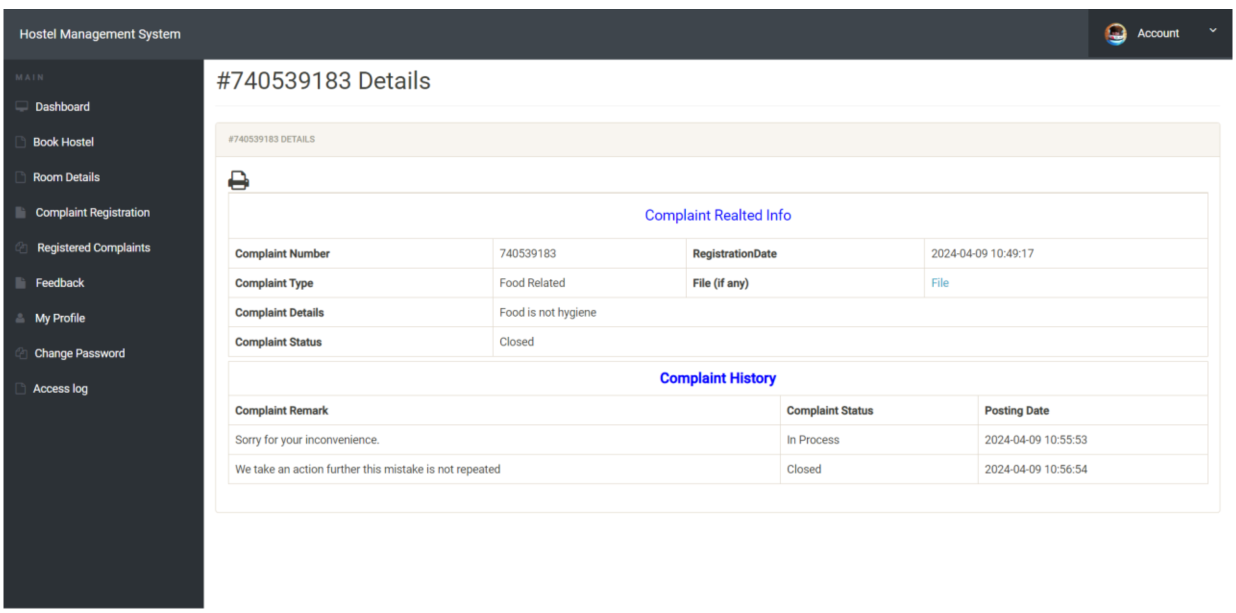
Complaint Form



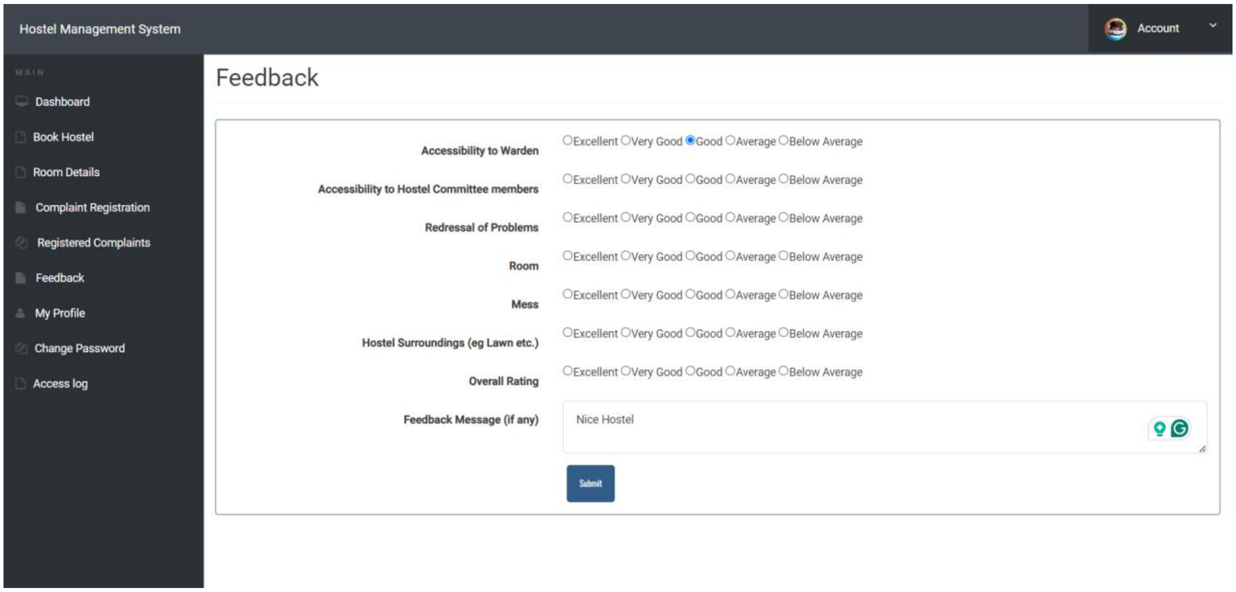
Complaint Lodges



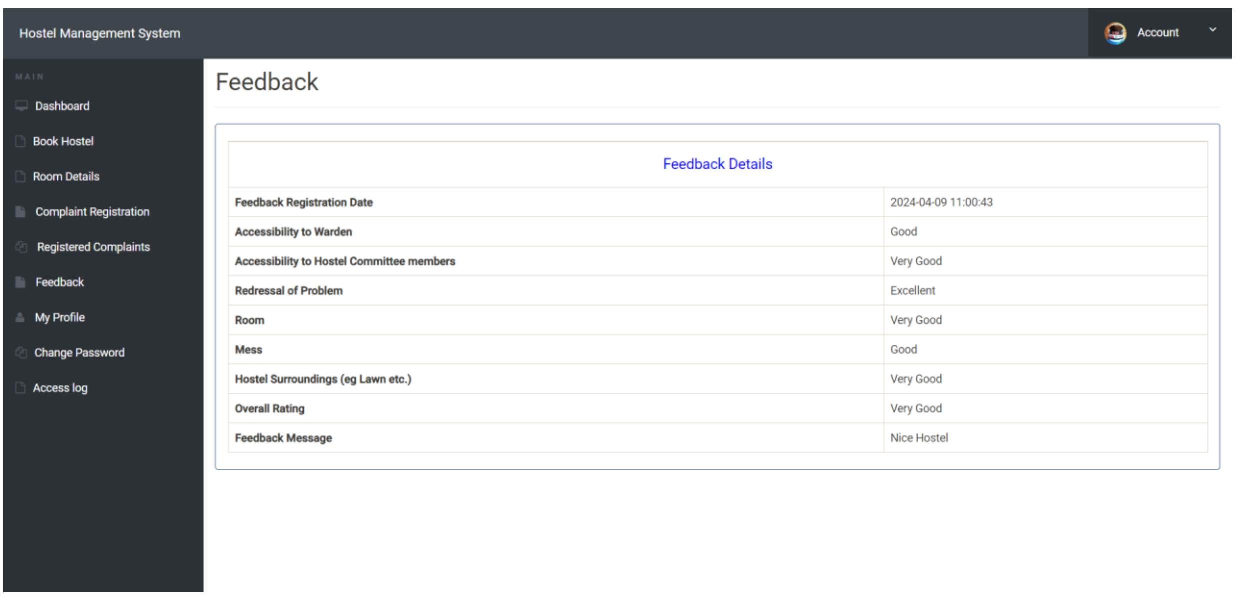
Details of Complaint



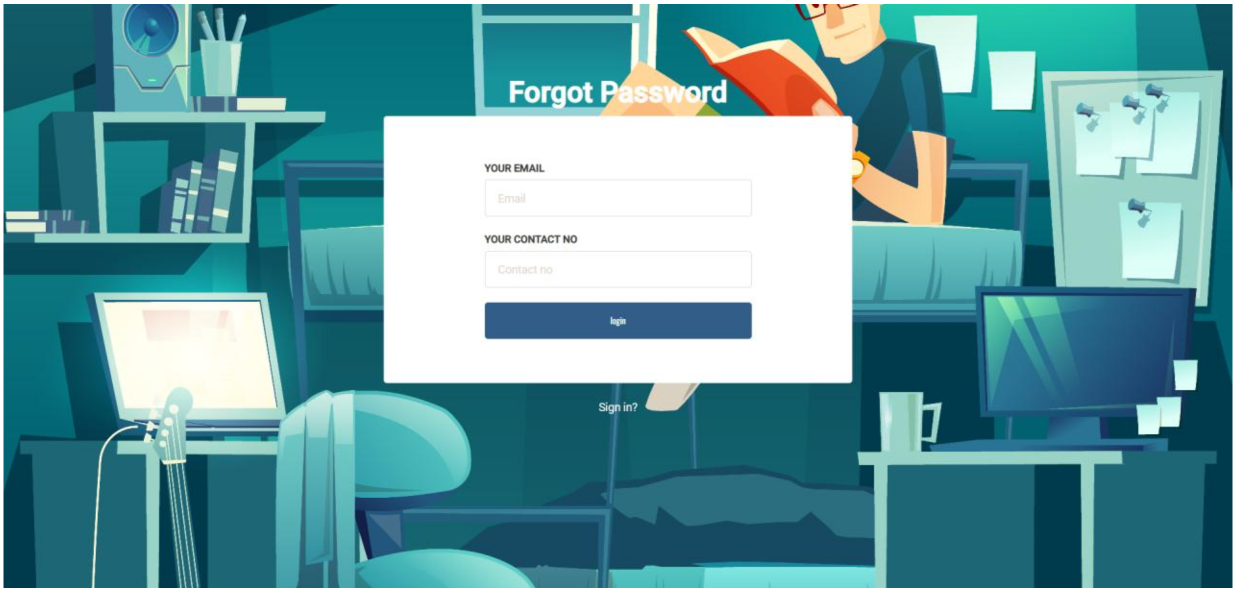
Feedback Form



Feedback Details

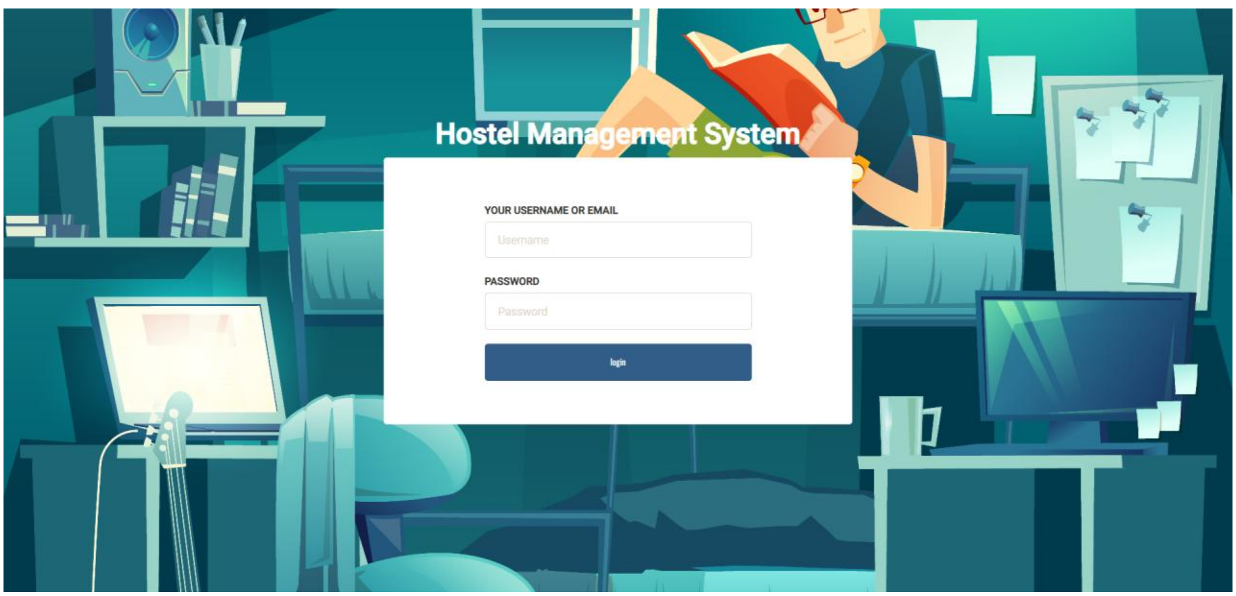


Forgot Password

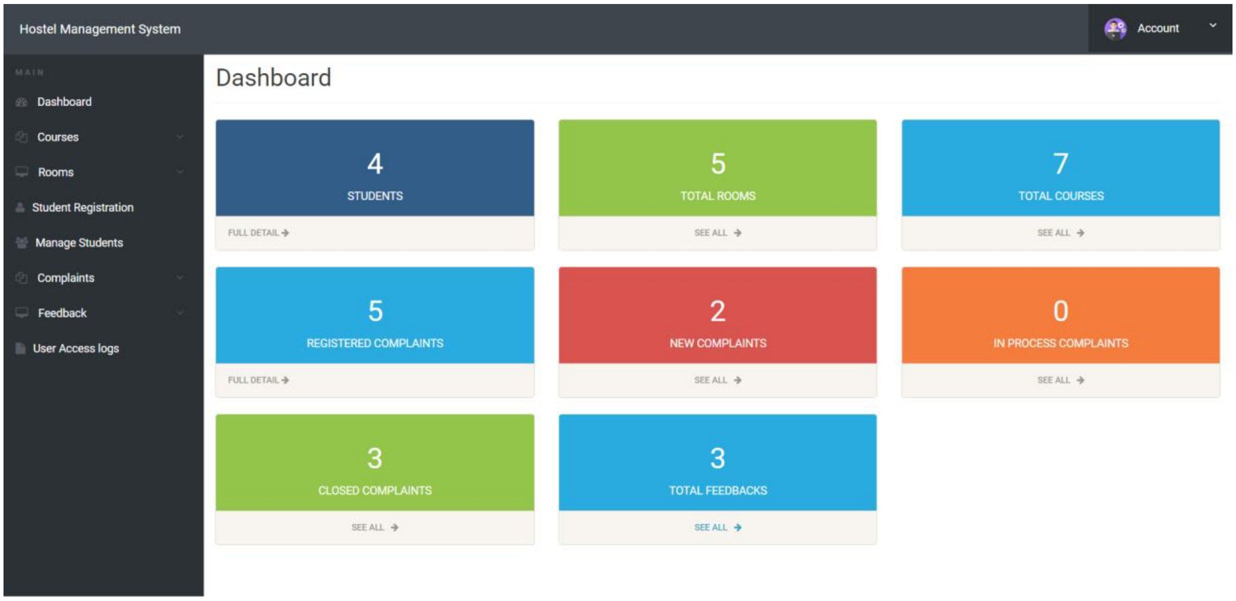


Admin Panel

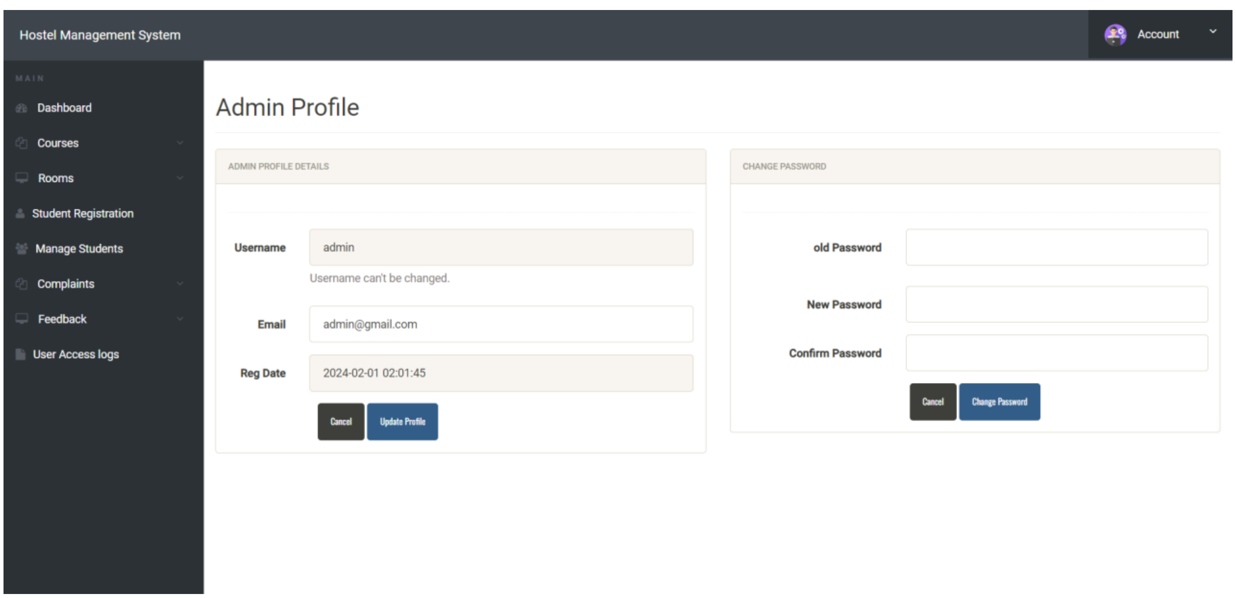
Login Page



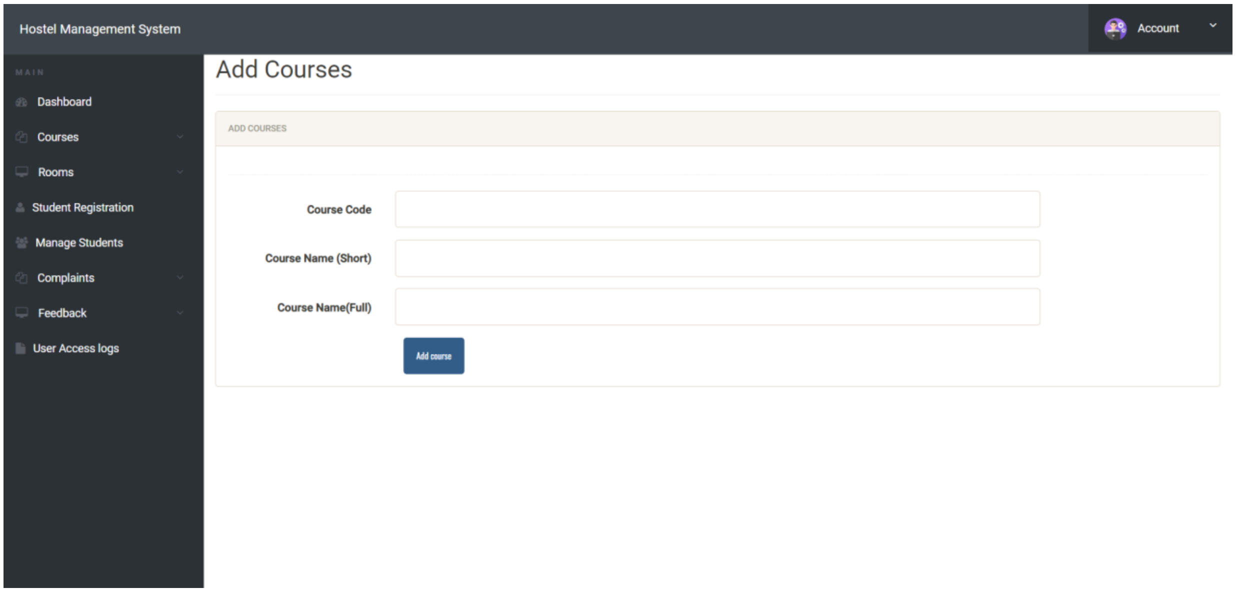
Dashboard



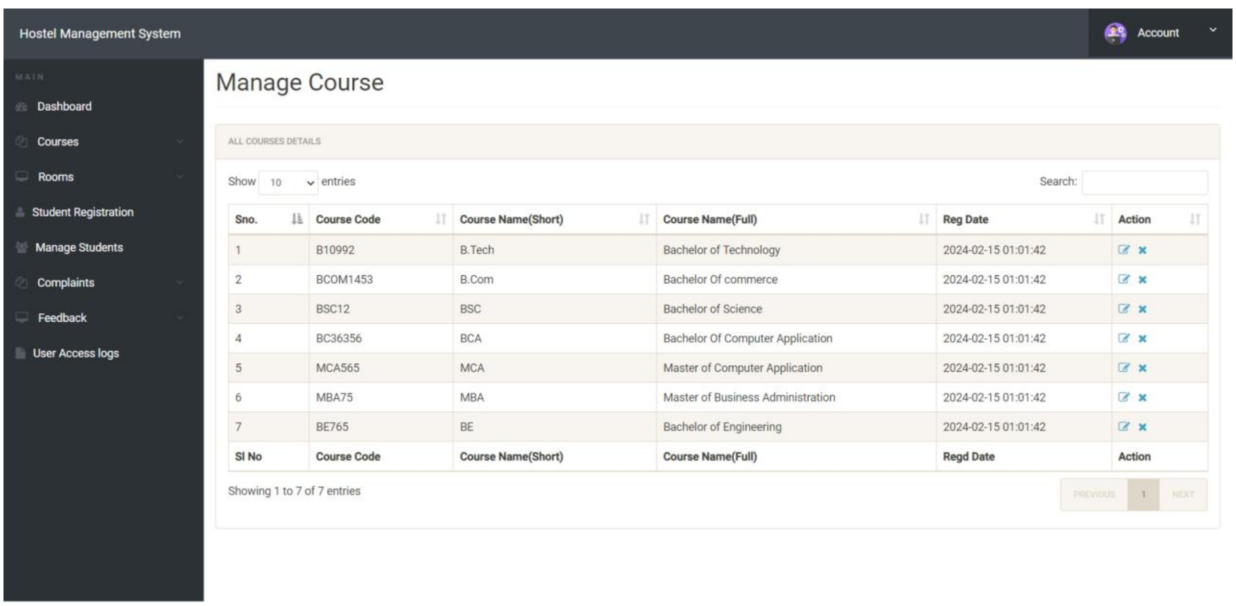
My Accounts



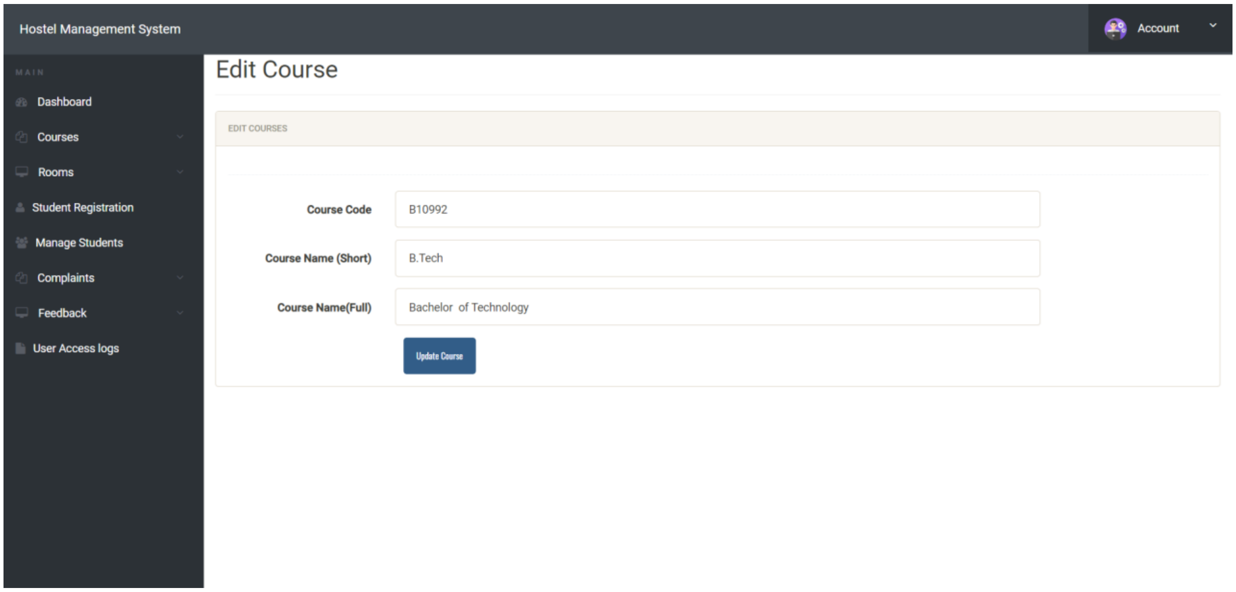
Add Courses



Manage Courses

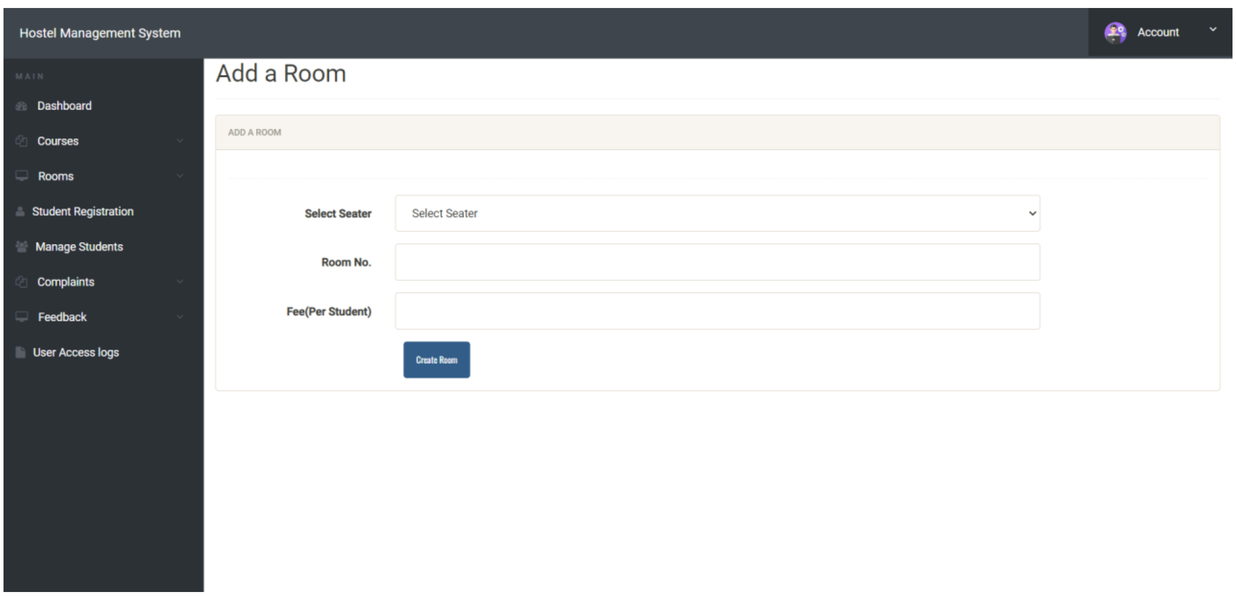


Update Courses



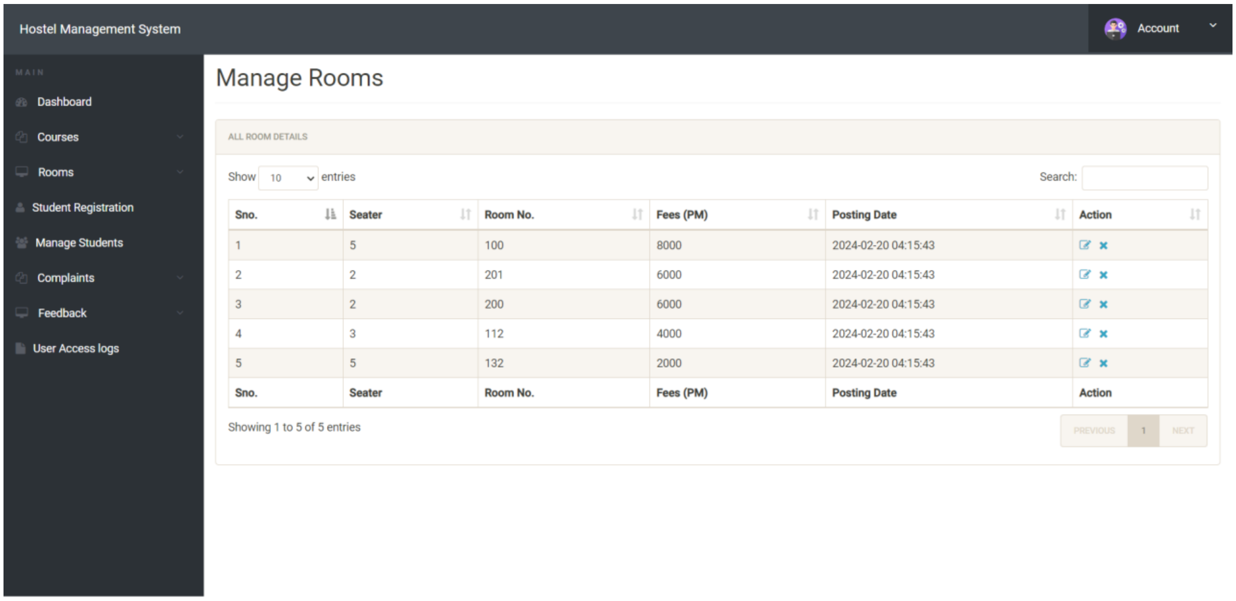
Hostel Management System
Account
MAIN
Dashboard
Courses
Rooms
Student Registration
Manage Students
Complaints
Feedback
User Access logs
Edit Course
EDIT COURSES
Course Code
B10992
Course Name (Short)
B.Tech
Course Name(Full)
Bachelor of Technology
Update Course

Add Room

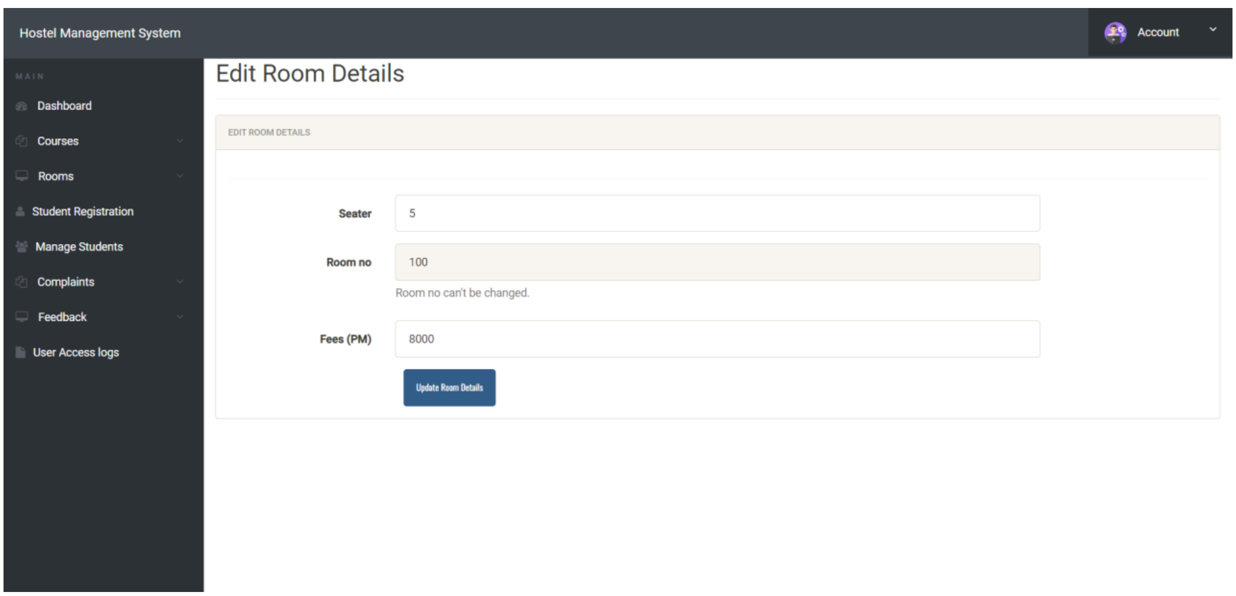


Hostel Management System
Account
MAIN
Dashboard
Courses
Rooms
Student Registration
Manage Students
Complaints
Feedback
User Access logs
Add a Room
ADD A ROOM
Select Seater
Select Seater
Room No.
Fee(Per Student)
Create Room

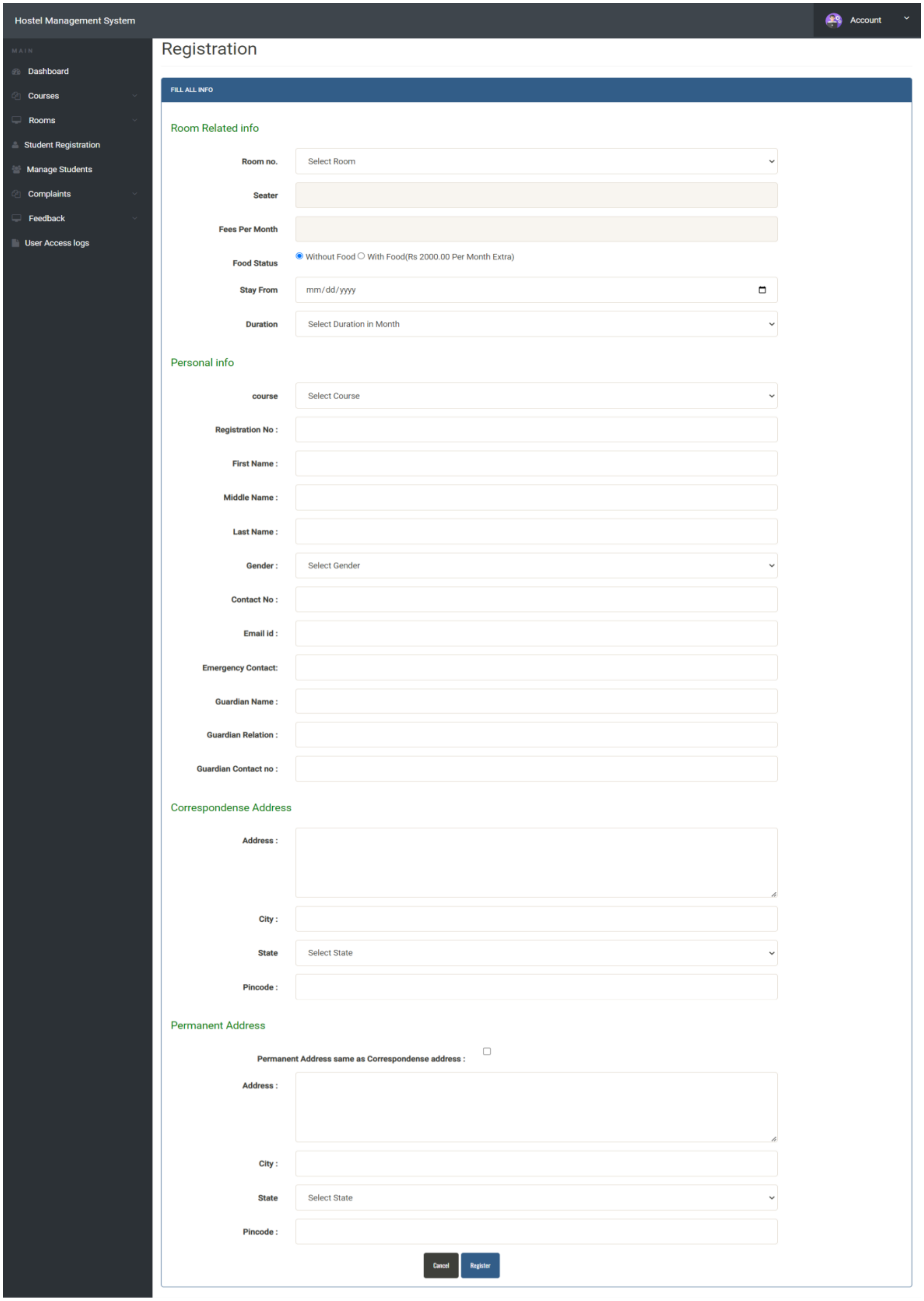
Manage Rooms



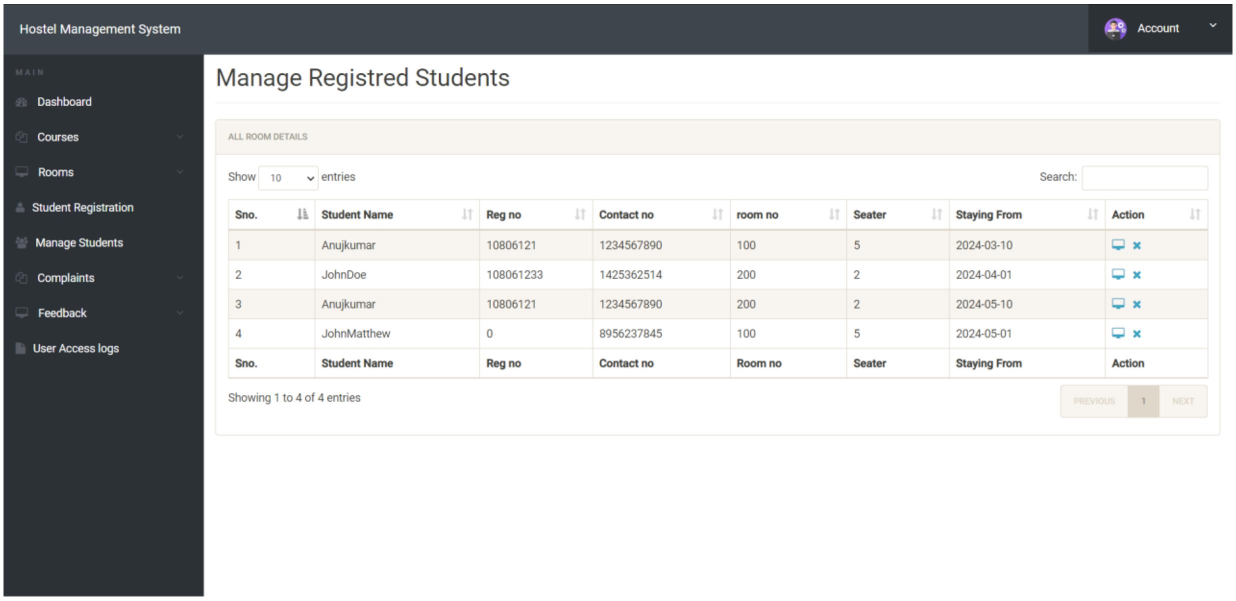
Update Room Details



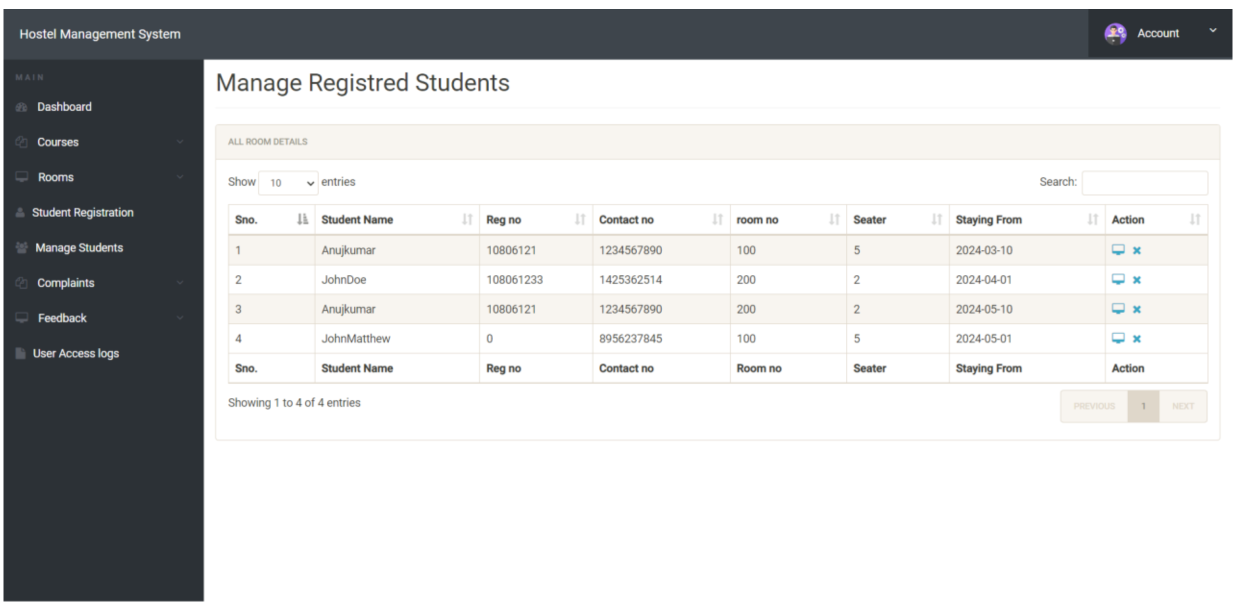
Student Registration


Hostel Management System
Account
MAIN
Dashboard
Courses
Rooms
Student Registration
Manage Students
Complaints
Feedback
User Access logs
Registration
FILL ALL INFO
Room Related info
Room no.
Select Room
Seater
Fees Per Month
Food Status
Without Food
With Food(Rs 2000.00 Per Month Extra)
Stay From
mm/dd/yyyy
Duration
Select Duration in Month
Personal info
course
Select Course
Registration No :
First Name :
Middle Name :
Last Name :
Gender :
Select Gender
Contact No :
Email id :
Emergency Contact:
Guardian Name :
Guardian Relation :
Guardian Contact no :
Correspondense Address
Address :
City :
State
Select State
Pincode :
Permanent Address
Permanent Address same as Correspondense address :
Address :
City :
State
Select State
Pincode :
Cancel
Register

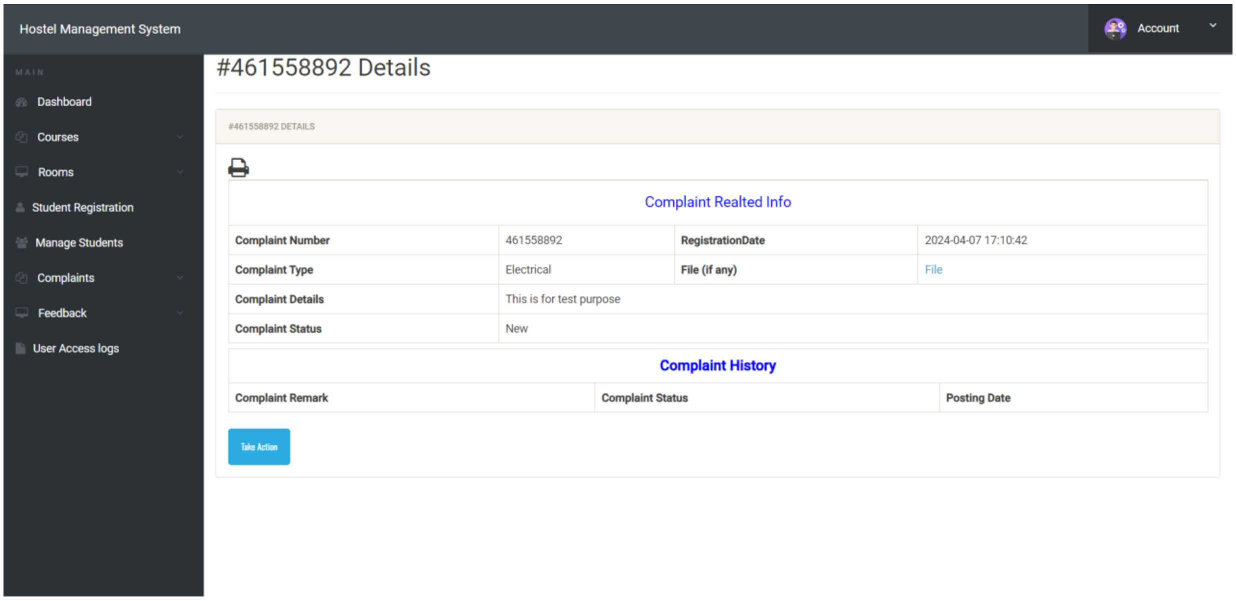
Manage Students



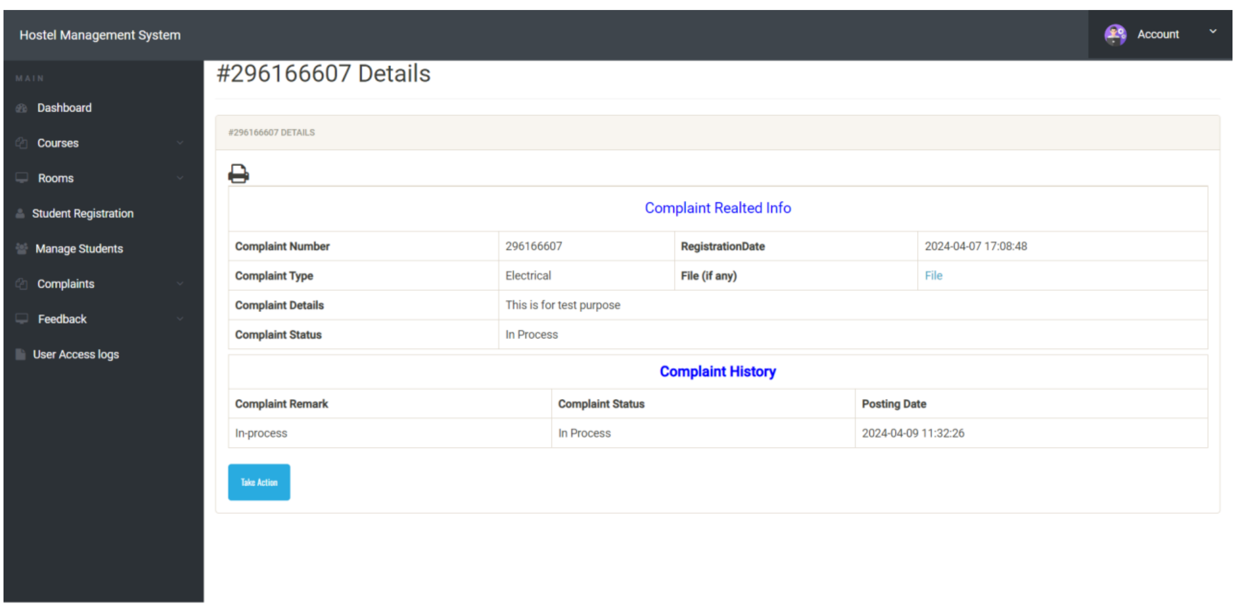
All Complaints



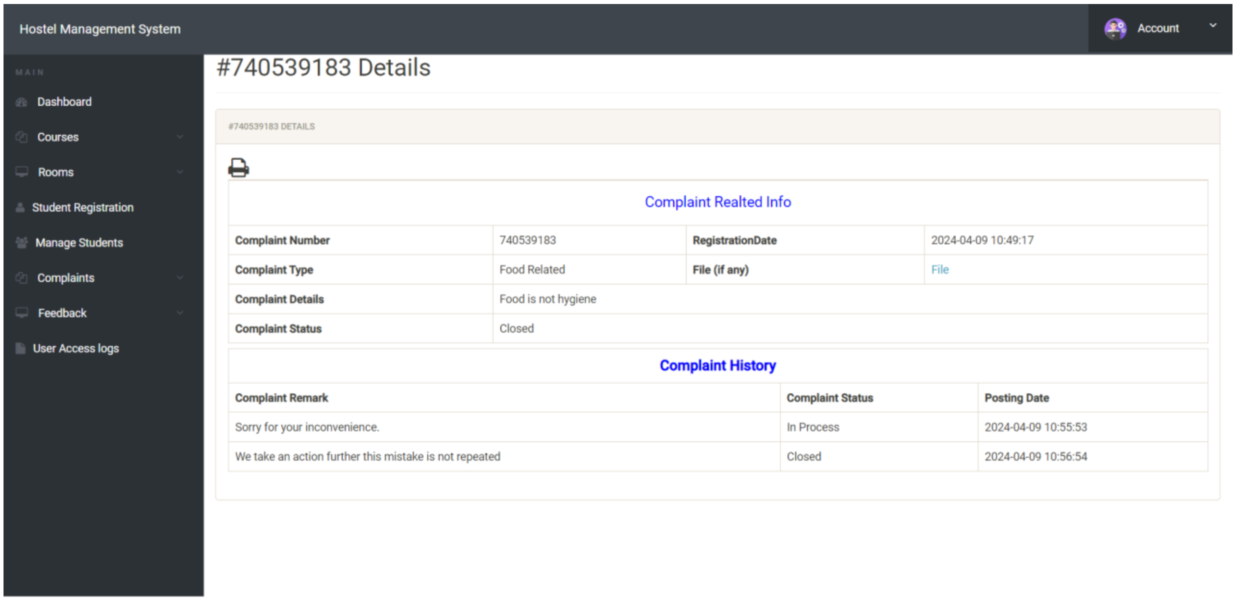
New Complaints Details



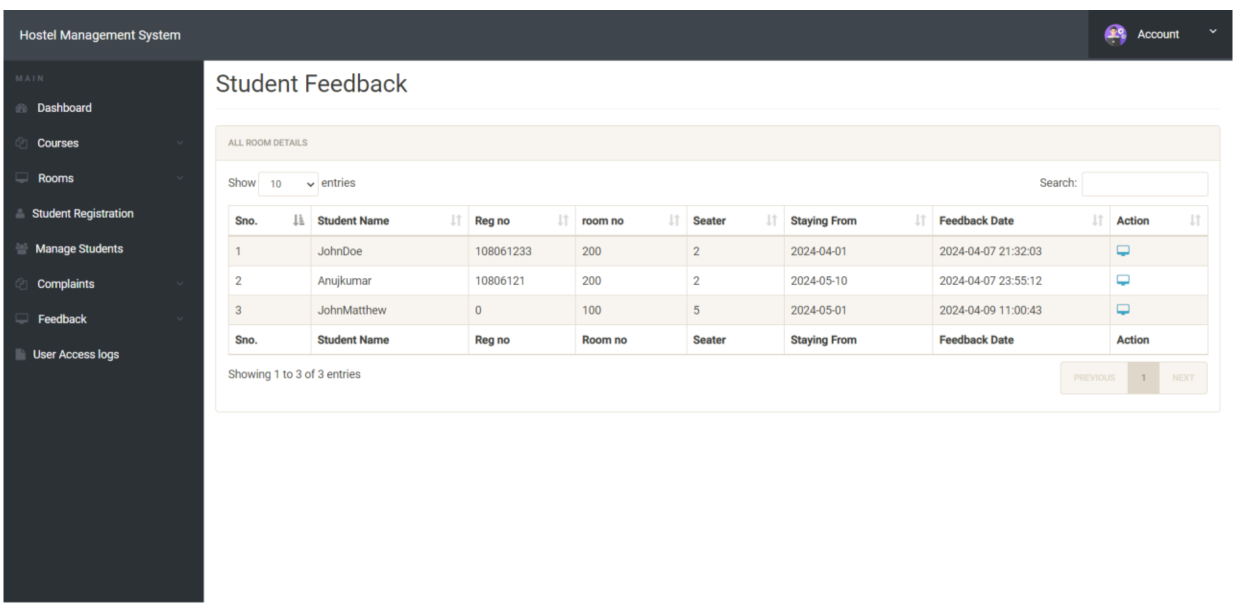
In process complaints Details



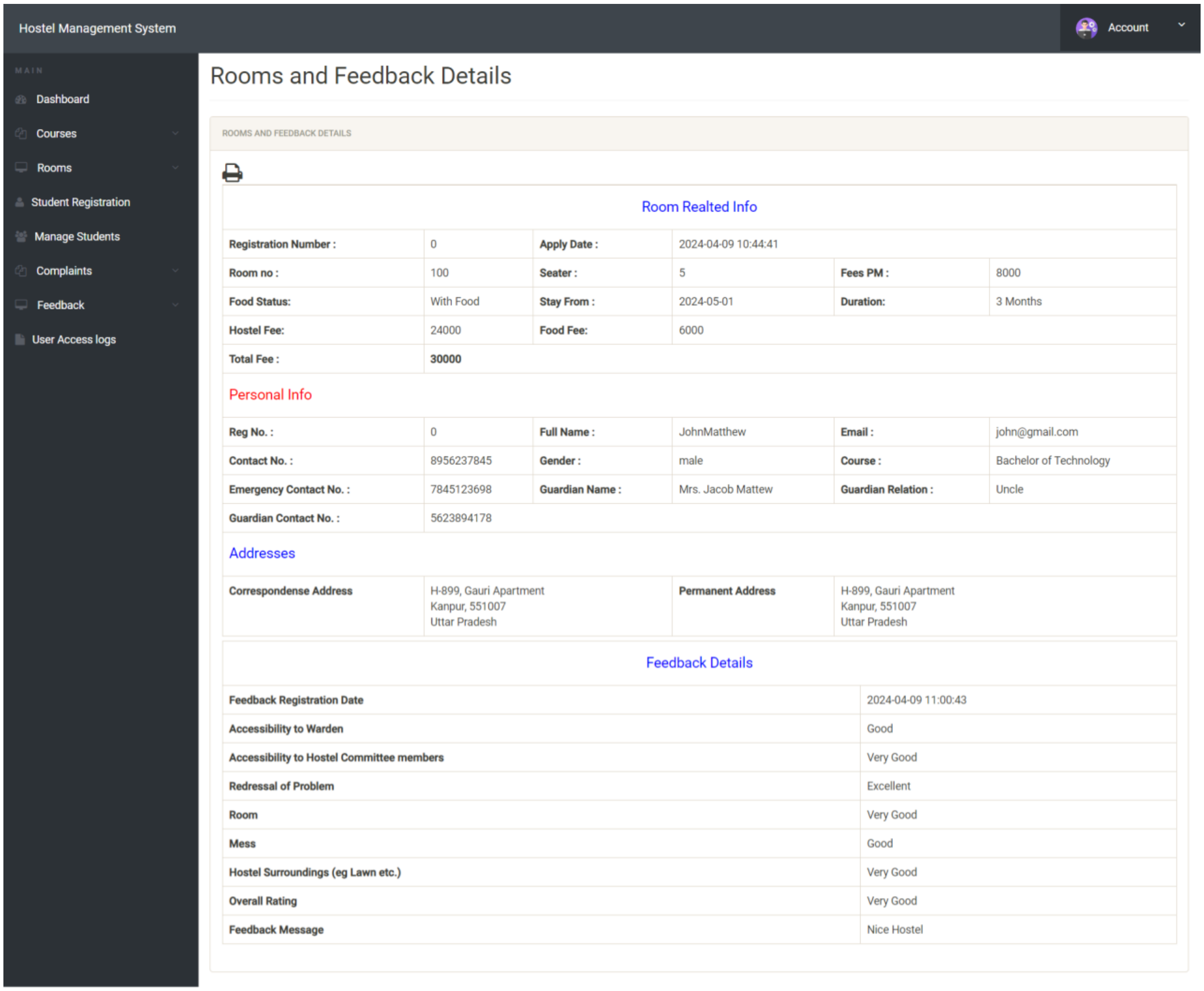
Closed Complaints Details



Student Feedback



Rooms and Feedback Details



Hostel Management System

Account

MAIN

- Dashboard
- Courses
- Rooms
- Student Registration
- Manage Students
- Complaints
- Feedback
- User Access logs

Rooms and Feedback Details

ROOMS AND FEEDBACK DETAILS

Room Realted Info

Registration Number :	0	**Apply Date :**	2024-04-09 10:44:41		
Room no :	100	**Seater :**	5	**Fees PM :**	8000
Food Status:	With Food	**Stay From :**	2024-05-01	**Duration:**	3 Months
Hostel Fee:	24000	**Food Fee:**	6000		
Total Fee :	**30000**				

Personal Info

Reg No. :	0	**Full Name :**	JohnMatthew	**Email :**	john@gmail.com
Contact No. :	8956237845	**Gender :**	male	**Course :**	Bachelor of Technology
Emergency Contact No. :	7845123698	**Guardian Name :**	Mrs. Jacob Mattew	**Guardian Relation :**	Uncle
Guardian Contact No. :	5623894178				

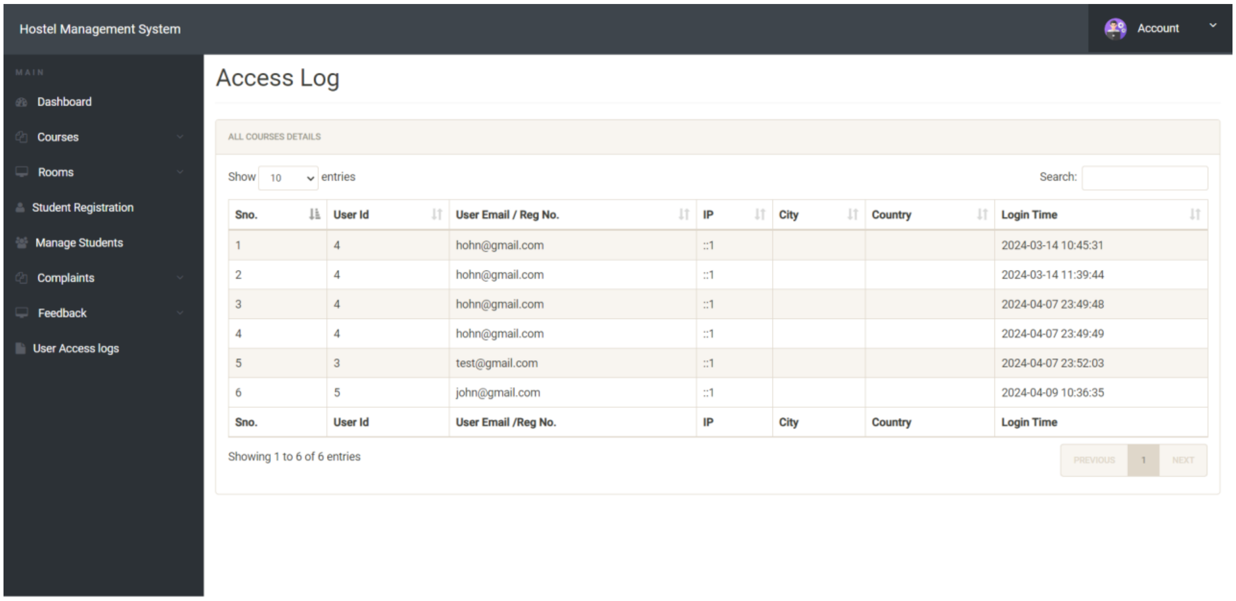
Addresses

Correspondense Address	H-899, Gauri Apartment Kanpur, 551007 Uttar Pradesh	**Permanent Address**	H-899, Gauri Apartment Kanpur, 551007 Uttar Pradesh

Feedback Details

Feedback Registration Date	2024-04-09 11:00:43
Accessibility to Warden	Good
Accessibility to Hostel Committee members	Very Good
Redressal of Problem	Excellent
Room	Very Good
Mess	Good
Hostel Surroundings (eg Lawn etc.)	Very Good
Overall Rating	Very Good
Feedback Message	Nice Hostel

User Access Log



CONCLUSION

To conclude the description about the project: The project, developed using PHP and MySQL is based on the requirement specification of the user and the analysis of the existing system, with flexibility for future enhancement.

The expanded functionality of today’s software requires an appropriate approach towards software development. This hostel management software is designed for people who want to manage various activi-ties in the hostel. For the past few years the number of educational institutions are increasing rapidly.

Thereby the number of hostels are also increasing for the accommodation of the students studying in this institution. And hence there is a lot of strain on the person who are running the hostel and software’s are not usually used in this context. This particular project deals with the problems on managing a hostel and avoids the problems which occur when carried manually.

Identification of the drawbacks of the existing system leads to the designing of computerized system that will be compatible to the existing system with the system which is more user friendly and more GUI oriented.

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